FIT PM12 twelve-channel test campaign

FIT hardware acceptance bench

November 13, 2026

1 Scope and recorded coverage

Board PM12-K7-3A17A1B402212223. Campaign 5ef2e2b4-c04a-4fad-b86f-a23be621d7a7.

Recorded status: INCOMPLETE. Missing tests are NOT MEASURED. Complete selected-protocol coverage is not a calibrated whole-board release.

Channel	cfd-monitor	cfd-dacs	gate-phase	tdc	adc-dc	adc-lines	integrator
1	NM	NM	NM	NM	NM	NM	NM
2	NM	NM	NM	NM	NM	NM	NM
3	NM	NM	NM	NM	NM	NM	NM
4	NM	NM	NM	NM	NM	NM	NM
5	NM	NM	NM	NM	NM	NM	NM
6	NM	NM	NM	NM	NM	NM	NM
7	NM	NM	NM	NM	NM	NM	NM
8	NM	NM	NM	NM	NM	NM	NM
9	NM	PASS	PASS	PASS	NM	NM	NM
10	NM	PASS	PASS	PASS	PASS	PASS	PASS
11	NM	PASS	PASS	PASS	PASS	PASS	PASS
12	NM	PASS	PASS	PASS	PASS	PASS	PASS

NM = not measured; INC = incomplete; ERR = acquisition error. No missing series is invented.

1.1 Board-level tests

Test	Status
Boundary scan and CPLD–FPGA connections	NOT_MEASURED
MCU, CPU–FPGA communication and peripheral IDs	NOT_MEASURED
I2C sensors, currents and voltages	NOT_MEASURED
HDMI digital link	NOT_MEASURED
Ethernet/bridge diagnostic	NOT_MEASURED

2 Fitted batches and fixture provenance

Batch attempt	PM channel	CFD label	Declared connection	Fitting
ch09-12-attempt001	12	CFD01	slot 7, RJ45 0	Identity/fitting confirmed
ch09-12-attempt001	11	CFD02	slot 7, RJ45 1	Identity/fitting confirmed
ch09-12-attempt001	10	CFD03	slot 7, RJ45 2	Identity/fitting confirmed
ch09-12-attempt001	9	CFD04	slot 7, RJ45 3	Identity/fitting confirmed

Fixture labels CFD01–CFD04 are provisional bench labels. Factory identities and photos remain unknown until supplied; a label is not proof of module identity. Declared cable routes must be checked physically.

Raw inputs and selected result IDs are retained in inputs/. Repeated measurements are never silently substituted.

3 Retained attempts and explicit selection

Result ID prefix	Test	Channel	Fitting	Status	Selected
result-b9126235	adc-dc	12	ch09-12-attempt001	PASS	yes
result-ee0613ea	adc-lines	12	ch09-12-attempt001	PASS	yes
result-217f5cb4	integrator	12	ch09-12-attempt001	PASS	yes
result-b99efe3a	cfd-dacs	9	ch09-12-attempt001	INCOMPLETE	no
result-ef50493e	cfd-dacs	10	ch09-12-attempt001	INCOMPLETE	no
result-c4800a66	cfd-dacs	11	ch09-12-attempt001	INCOMPLETE	no
result-c80f3b9e	cfd-dacs	12	ch09-12-attempt001	INCOMPLETE	no
result-9e64b43e	cfd-dacs	9	ch09-12-attempt001	PASS	yes
result-0fcaadd7	cfd-dacs	10	ch09-12-attempt001	PASS	yes
result-8b6b8aa7	cfd-dacs	11	ch09-12-attempt001	PASS	yes
result-b4c6c497	cfd-dacs	12	ch09-12-attempt001	PASS	yes
result-aa2f2b7c	adc-dc	11	ch09-12-attempt001	PASS	yes
result-fa68f809	adc-lines	11	ch09-12-attempt001	PASS	yes
result-021f9e97	integrator	11	ch09-12-attempt001	PASS	yes
result-317b8d60	adc-dc	10	ch09-12-attempt001	PASS	yes

Result ID prefix	Test	Channel	Fitting	Status	Selected
result-0386d766	adc-lines	10	ch09-12-attempt001	PASS	yes
result-e87c1c5e	integrator	10	ch09-12-attempt001	PASS	yes
result-1642153e	tdc	12	ch09-12-attempt001	PASS	yes
result-5f833ba0	tdc	11	ch09-12-attempt001	PASS	yes
result-cdf2341e	tdc	10	ch09-12-attempt001	PASS	yes
result-5e3a4c64	tdc	9	ch09-12-attempt001	PASS	yes
result-d2e5cd6c	gate-phase	12	ch09-12-attempt001	PASS	yes
result-769a5717	gate-phase	11	ch09-12-attempt001	PASS	yes
result-5b389adb	gate-phase	10	ch09-12-attempt001	PASS	yes
result-a00721bb	gate-phase	9	ch09-12-attempt001	PASS	yes

All attempts, including failures and rejected repeats, remain in the archived inputs. The coverage matrix uses only the explicitly selected attempt.

4 Selected measurement conditions and limits

Test	Physical channel	Confirmed stimulus connection
adc-dc	12	{”dac_channels”: [0, 1], ”stimulus_channel”: 12, ”usb_input”: ”J8”}
adc-lines	12	{”dac_channels”: [0, 1], ”stimulus_channel”: 12, ”usb_input”: ”J8”}
integrator	12	{”dac_channels”: [2, 3], ”stimulus_channel”: 12, ”usb_input”: ”J6”}
cfd-dacs	9	{”selection”: ”one monitor mux enabled; all others inhibited”, ”shared_monitor_channels”: [9, 10, 11, 12]}
cfd-dacs	10	{”selection”: ”one monitor mux enabled; all others inhibited”, ”shared_monitor_channels”: [9, 10, 11, 12]}
cfd-dacs	11	{”selection”: ”one monitor mux enabled; all others inhibited”, ”shared_monitor_channels”: [9, 10, 11, 12]}

Test	Physical channel	Confirmed stimulus connec tion
cfd-dacs	12	{”selection”: ”one monitor mux enabled; all others inhibited”, ”shared_monito r_channels”: [9, 10, 11, 12]}
adc-dc	11	{”dac_channels”: [0, 1], ”stimulus_channel”: 11, ”usb_input”: ”J8”}
adc-lines	11	{”dac_channels”: [0, 1], ”stimulus_channel”: 11, ”usb_input”: ”J8”}
integrator	11	{”dac_channels”: [2, 3], ”stimulus_channel”: 11, ”usb_input”: ”J6”}
adc-dc	10	{”dac_channels”: [0, 1], ”stimulus_channel”: 10, ”usb_input”: ”J8”}
adc-lines	10	{”dac_channels”: [0, 1], ”stimulus_channel”: 10, ”usb_input”: ”J8”}
integrator	10	{”dac_channels”: [2, 3], ”stimulus_channel”: 10, ”usb_input”: ”J6”}
tdc	12	{”phase_lanes”: [8, 9], ”rj45_port”: 0, ”slot”: 7, ”stimulus_dio”: 0}
tdc	11	{”phase_lanes”: [10, 11], ”rj45_port”: 1, ”slot”: 7, ”stimulus_dio”: 4}
tdc	10	{”phase_lanes”: [12, 13], ”rj45_port”: 2, ”slot”: 7, ”stimulus_dio”: 8}
tdc	9	{”phase_lanes”: [14, 15], ”rj45_port”: 3, ”slot”: 7, ”stimulus_dio”: 12}
gate-phase	12	{”phase_lanes”: [8, 9], ”rj45_port”: 0, ”slot”: 7, ”stimulus_dio”: 0}
gate-phase	11	{”phase_lanes”: [10, 11], ”rj45_port”: 1, ”slot”: 7, ”stimulus_dio”: 4}
gate-phase	10	{”phase_lanes”: [12, 13], ”rj45_port”: 2, ”slot”: 7, ”stimulus_dio”: 8}
gate-phase	9	{”phase_lanes”: [14, 15], ”rj45_port”: 3, ”slot”: 7, ”stimulus_dio”: 12}

4.1 Common hardware configuration

Setting	Recorded value
clock_profile.adc_rate	2x reference
clock_profile.kasli_rate	3x reference
clock_profile.reference_mhz	40.0787
cpld_gate_mode	original INTA/INTB enabled, nomina l20MHz
kasli_build.ABI	CFD4
kasli_build.expected_bit_sha256	a71a08a258620b0e5a52c5d03aefd8c69040 c9d15e8de6391c484ac182b7173b
pm_build	c7c42d28

4.2 adc-dc

Setting	Recorded value
method	”direct-ADC-DC-v1”
stimulus_points_per_bank	257
commanded_range_V	[-1.2, 1.2]
common_mode_V	1.0
dac_channels	[0, 1]
fit_interval_V	[-0.8, 0.8]
fft_sample_rate_assumed_Hz	80157400
integrator_input_parked_V	[2.0480001, 1.02400005]

4.3 adc-lines

Setting	Recorded value
method	”adjacent-pair-states-v1”
stimulus_points_per_bank	257
commanded_range_V	[-1.2, 1.2]
common_mode_V	1.0
dac_channels	[0, 1]
fit_interval_V	[-0.8, 0.8]
fft_sample_rate_assumed_Hz	80157400
integrator_input_parked_V	[2.0480001, 1.02400005]

4.4 cfd-dacs

Setting	Recorded value
method	”shared-monitor-SMU-BCD-v1”
instrument_identity	”Keysight Technologies,B2910BL,MY6332 0299,5.0.2429.8966”
sense	”Force HI/LO, local”
source_current_A	0
source_range_A	1e-06
voltage_range_V	20
nplc	1
settling_s	0.1
points	101
repeats	3

Setting	Recorded value
limits	{"repeat_std_V": 0.001, "reverse_step_V": 0.0002, "span_relative_error": 0.05}
post_scan_verification	"SMU OFF/HIZ, four monitor muxes inhibited, explicit TCA masks0"

4.5 gate-phase

Setting	Recorded value
method	"DAC-A-ISERDES-gate-phase-v1"
codes	[0, 655, 1311, 1966, 2621, 3277, 3932, 4587, 5243, 5898, 6554, 7209, 7864, 8520, 9175, 9830, 10486, 11141, 11796, 12452, 13107, 13762, 14418, 15073, 15728, 16384, 17039, 17694, 18350, 19005, 19661, 20316, 20971, 21627, 22282, 22937, 23593, 24248, 24903, 25559, 26214, 26869, 27525, 28180, 28835, 29491, 30146, 30801, 31457, 32112, 32768, 33423, 34078, 34734, 35389, 36044, 36700, 37355, 38010, 38666, 39321, 39976, 40632, 41287, 41942, 42598, 43253, 43908, 44564, 45219, 45875, 46530, 47185, 47841, 48496, 49151, 49807, 50462, 51117, 51773, 52428, 53083, 53739, 54394, 55049, 55705, 56360, 57015, 57671, 58326, 58982, 59637, 60292, 60948, 61603, 62258, 62914, 63569, 64224, 64880, 65535]
sample_hz	961888800.0
gate_period_bins	48
capture_words	4095
minimum_functional_excursion_ns	5

4.6 integrator

Setting	Recorded value
method	"signed-integration-pair-v1"
dac_channels	[2, 3]
base_V	[1.024, 2.048]
coarse_offsets_V	[-0.09999999999999987, -0.08000000000000007, -0.05999999999999983, -0.04000000000000036, -0.019999999999999796, -0.010000000000000009, 0.0, 0.009999999999999787, 0.02000000000000024, 0.04000000000000036, 0.060000000000000275, 0.08000000000000007, 0.10000000000000009]
fine_method	"26 points,0.4mV step around coarse estimate"
return_method	"two candidate revisits and +/-5mV offsets;near-zero <=2codes; signed response >4codes"

4.7 tdc

Setting	Recorded value
method	”finite-OSERDES-TDC-phase-v1”
phase_slots	[24, 25, 26, 27, 28, 29, 30, 31, 32, 33, 34, 35, 36, 37, 38, 39, 40, 41, 42, 43, 44, 45, 46, 47, 48]
sample_hz	961888800.0
events_per_phase	128
discovery_events	16
response_slope_window	[-60, -20]
maximum_residual_fine_codes	10

Commanded DAC voltages, whole connected chain; not calibrated ADC-only INL/noise or electrical continuity proof.

Effective differential whole-chain balance; waveform-based integration pair, not an absolute gate timestamp or calibrated opamp bias.

Fine modulo128 unwrapped by nearest continuation; coarse fields retained without repair. Progra mmed slot1.0396ns assumes40.0787MHz; not calibrated TDC timing or intrinsic jitter.

Functional assembled-chain transfer check: span within5% of schematic scaling, reverse step <=0.2mV, repeat standard deviation <=1mV. Not traceable calibration or exhaustive16-bit INL/DNL. DAC-A unchanged; exercised B/C/D end at0000..

Occupancy-weighted sampled phase centroid; nominal1.0396ns bins and49.9ns gate period. Neares t-continuation unwrap; discontinuities retained. No calibrated sub-bin precision or independent-edge sqrt(N) uncertainty. DAC A commanded0000 at cleanup, without hardware code readback.

integrator; physical channel 10: The coarse waveform assignment produces an additional positive-going sign crossing near +40mV. It was not accepted as balance. Fine acquisition used the central negati ve-going crossing for this reversed input coordinate, then confirmed it with independent candidate revisi ts and +/-5mV controls. The original coarse data and both controller versions remain archived.

ADC total RMS includes the complete connected chain and periodic gate feedthrough. The four-phase-removed RMS removes the mean of each sample phase; it is a residual statistic, not an intrinsic ADC noise specification. FFT and time axes use the declared clock frequency, not an independent frequency measurement.

5 Measurements and common channel plots

Test	Channel	Bank	Observable	Value	Unit
adc-dc	12	A	gain	-1784.205440671235	code/V
adc-dc	12	A	zero_mean	2162.0811339198435	code
adc-dc	12	A	zero_total_rms	56.22754588987761	code
adc-dc	12	A	zero_four_phase_removed_rms	0.5926371250728474	code
adc-dc	12	B	gain	-1784.2694869545635	code/V
adc-dc	12	B	zero_mean	2165.8475073313784	code

Test	Channel	Bank	Observable	Value	Unit
adc-dc	12	B	zero_total_rms	64.40894678040004	code
adc-dc	12	B	zero_four_phase_removed_rms	0.5625571866355403	code
adc-lines	12	A	adjacent_pair_states	44	state
adc-lines	12	A	visited_codes	3147	code
adc-lines	12	B	adjacent_pair_states	44	state
adc-lines	12	B	visited_codes	3184	code
integrator	12	A	balance_delta	0.006030008721686399	V
integrator	12	A	gain	-2530.411960665549	code/V
integrator	12	A	near_balance_change	0.13016491045236994	code
integrator	12	B	balance_delta	0.007424191555263186	V
integrator	12	B	gain	-2448.0077895617496	code/V
integrator	12	B	near_balance_change	0.11194322217075807	code
cfd-dacs	9		dac_B_minimum_code_V	1.0447266666666666	V
cfd-dacs	9		dac_B_maximum_code_V	1.0046133333333334	V
cfd-dacs	9		dac_B_span_V	-0.04011333333333322	V
cfd-dacs	9		dac_B_maximum_reverse_step_V	0	V
cfd-dacs	9		dac_B_maximum_repeat_std_V	7.023769168570249e-05	V
cfd-dacs	9		dac_B_interior_gain_error_percent	-0.006716857635891138	%
cfd-dacs	9		dac_C_minimum_code_V	1.0349566666666667	V
cfd-dacs	9		dac_C_maximum_code_V	1.0143966666666666	V
cfd-dacs	9		dac_C_span_V	-0.020560000000000134	V
cfd-dacs	9		dac_C_maximum_reverse_step_V	0	V

Test	Channel	Bank	Observable	Value	Unit
cfd-dacs	9		dac_C_maximum_repeat_std_V	9.504384952928754e-05	V
cfd-dacs	9		dac_C_interior_gain_error_percent	0.8518971864797997	%
cfd-dacs	9		dac_D_minimum_code_V	0.0054	V
cfd-dacs	9		dac_D_maximum_code_V	1.06306	V
cfd-dacs	9		dac_D_span_V	1.0576599999999998	V
cfd-dacs	9		dac_D_maximum_reverse_step_V	0	V
cfd-dacs	9		dac_D_maximum_repeat_std_V	7.000000000000657e-05	V
cfd-dacs	9		dac_D_interior_gain_error_percent	-0.026880429186237897	%
cfd-dacs	10		dac_B_minimum_code_V	1.04437	V
cfd-dacs	10		dac_B_maximum_code_V	1.00438	V
cfd-dacs	10		dac_B_span_V	-0.03998999999999997	V
cfd-dacs	10		dac_B_maximum_reverse_step_V	0	V
cfd-dacs	10		dac_B_maximum_repeat_std_V	8.082903768656434e-05	V
cfd-dacs	10		dac_B_interior_gain_error_percent	-0.1422230480119024	%
cfd-dacs	10		dac_C_minimum_code_V	1.0346766666666667	V
cfd-dacs	10		dac_C_maximum_code_V	1.0141433333333334	V
cfd-dacs	10		dac_C_span_V	-0.020533333333292	V

Test	Channel	Bank	Observable	Value	Unit
cfd-dacs	10		dac_C_maximum_reverse_step_V	0	V
cfd-dacs	10		dac_C_maximum_repeat_std_V	9.53939201416794e-05	V
cfd-dacs	10		dac_C_interior_gain_error_percent	0.5828293358783654	%
cfd-dacs	10		dac_D_minimum_code_V	0.004546666666667	V
cfd-dacs	10		dac_D_maximum_code_V	1.0692933333333334	V
cfd-dacs	10		dac_D_span_V	1.0647466666667	V
cfd-dacs	10		dac_D_maximum_reverse_step_V	0	V
cfd-dacs	10		dac_D_maximum_repeat_std_V	7.000000000003828e-05	V
cfd-dacs	10		dac_D_interior_gain_error_percent	0.042824877925129456	%
cfd-dacs	11		dac_B_minimum_code_V	1.0446666666666	V
cfd-dacs	11		dac_B_maximum_code_V	1.00461	V
cfd-dacs	11		dac_B_span_V	-0.04005666666663	V
cfd-dacs	11		dac_B_maximum_reverse_step_V	0	V
cfd-dacs	11		dac_B_maximum_repeat_std_V	7.549834435276095e-05	V
cfd-dacs	11		dac_B_interior_gain_error_percent	-0.1743207749736131	%
cfd-dacs	11		dac_C_minimum_code_V	1.0349033333333333	V
cfd-dacs	11		dac_C_maximum_code_V	1.0143866666665	V

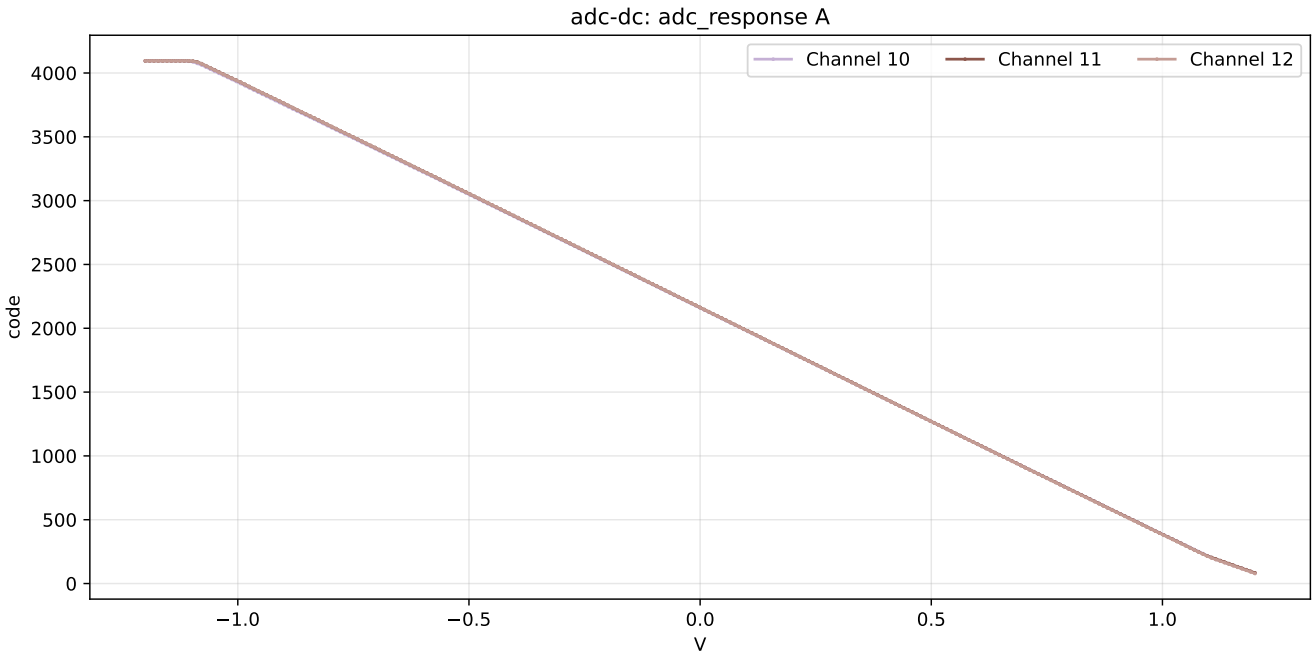
Test	Channel	Bank	Observable	Value	Unit
cfd-dacs	11		dac_C_span_V	-0.020516 66666666674	V
cfd-dacs	11		dac_C_maximum_reverse_step_V	0	V
cfd-dacs	11		dac_C_maximum_repeat_std_V	7.234178 138066471e-05	V
cfd-dacs	11		dac_C_interior_gain_error_percent	0.177204 94086375105	%
cfd-dacs	11		dac_D_minimum_code_V	0.00485	V
cfd-dacs	11		dac_D_maximum_code_V	1.063633 3333333334	V
cfd-dacs	11		dac_D_span_V	1.058783 3333333334	V
cfd-dacs	11		dac_D_maximum_reverse_step_V	0	V
cfd-dacs	11		dac_D_maximum_repeat_std_V	6.110100 926609052e-05	V
cfd-dacs	11		dac_D_interior_gain_error_percent	0.026777 441441017125	%
cfd-dacs	12		dac_B_minimum_code_V	1.04463	V
cfd-dacs	12		dac_B_maximum_code_V	1.004426 6666666666	V
cfd-dacs	12		dac_B_span_V	-0.040203 33333333337	V
cfd-dacs	12		dac_B_maximum_reverse_step_V	0	V
cfd-dacs	12		dac_B_maximum_repeat_std_V	4.932882 8623155544e-05	V
cfd-dacs	12		dac_B_interior_gain_error_percent	0.065108 73936658168	%
cfd-dacs	12		dac_C_minimum_code_V	1.034806 6666666667	V

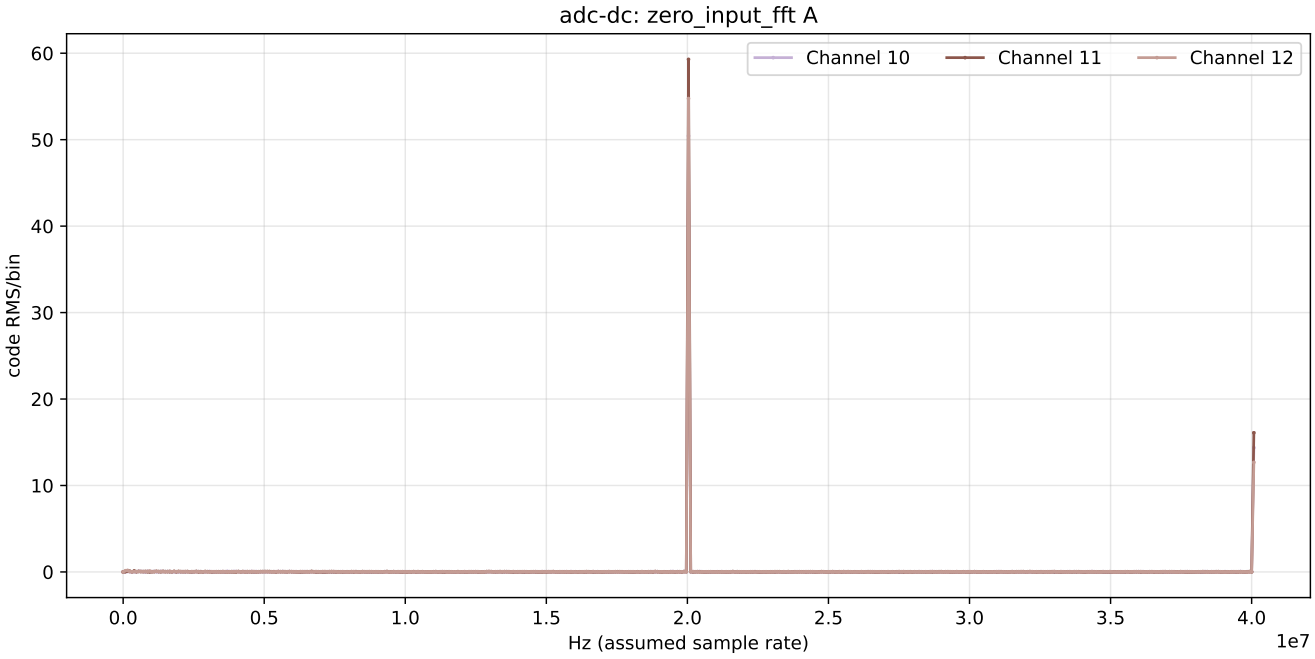
Test	Channel	Bank	Observable	Value	Unit
cfd-dacs	12		dac_C_maximum_code_V	1.0144333333333333	V
cfd-dacs	12		dac_C_span_V	-0.020373333333333354	V
cfd-dacs	12		dac_C_maximum_reverse_step_V	3.3333333335772153e-06	V
cfd-dacs	12		dac_C_maximum_repeat_std_V	0.00011846237095938458	V
cfd-dacs	12		dac_C_interior_gain_error_percent	-1.3950279374863153	%
cfd-dacs	12		dac_D_minimum_code_V	0.00498	V
cfd-dacs	12		dac_D_maximum_code_V	1.06321	V
cfd-dacs	12		dac_D_span_V	1.05823	V
cfd-dacs	12		dac_D_maximum_reverse_step_V	0	V
cfd-dacs	12		dac_D_maximum_repeat_std_V	8.020806277010361e-05	V
cfd-dacs	12		dac_D_interior_gain_error_percent	-0.03786962571341594	%
adc-dc	11	A	gain	-1783.3085324525614	code/V
adc-dc	11	A	zero_mean	2163.592375366569	code
adc-dc	11	A	zero_total_rms	61.45588679712764	code
adc-dc	11	A	zero_four_phase_removed_rms	0.5394259839466975	code
adc-dc	11	B	gain	-1776.8598609607993	code/V
adc-dc	11	B	zero_mean	2158.097751710655	code
adc-dc	11	B	zero_total_rms	59.53654688741656	code
adc-dc	11	B	zero_four_phase_removed_rms	0.5120989067289364	code
adc-lines	11	A	adjacent_pair_states	44	state

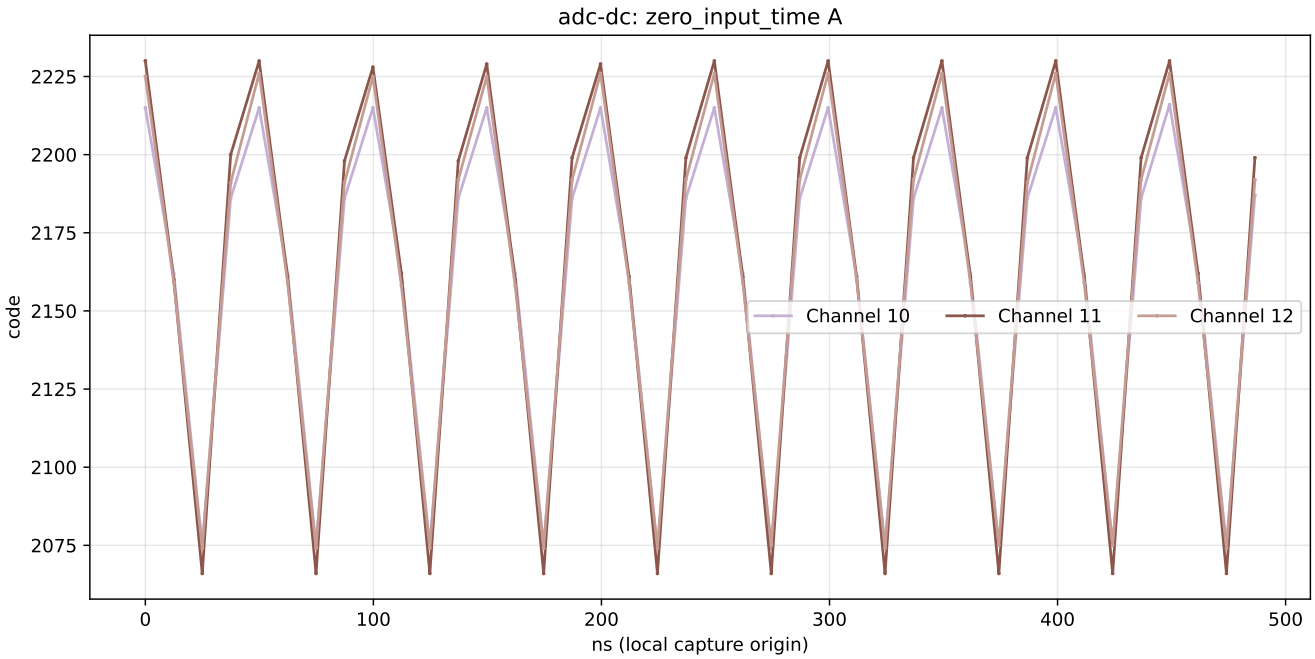
Test	Channel	Bank	Observable	Value	Unit
adc-lines	11	A	visited_codes	3175	code
adc-lines	11	B	adjacent_pair_states	44	state
adc-lines	11	B	visited_codes	3237	code
integrator	11	A	balance_delta	0.003963723409567792	V
integrator	11	A	gain	-2558.7140130296752	code/V
integrator	11	A	near_balance_change	-0.13615726802532035	code
integrator	11	B	balance_delta	0.009665397533032197	V
integrator	11	B	gain	-2553.295843675655	code/V
integrator	11	B	near_balance_change	0.1115210552725027	code
adc-dc	10	A	gain	-1777.6734930206578	code/V
adc-dc	10	A	zero_mean	2159.386119257087	code
adc-dc	10	A	zero_total_rms	52.46655577574308	code
adc-dc	10	A	zero_four_phase_removed_rms	0.5128491252775638	code
adc-dc	10	B	gain	-1778.2659098179804	code/V
adc-dc	10	B	zero_mean	2165.9296187683285	code
adc-dc	10	B	zero_total_rms	59.04859019865752	code
adc-dc	10	B	zero_four_phase_removed_rms	0.5082101785128094	code
adc-lines	10	A	adjacent_pair_states	44	state
adc-lines	10	A	visited_codes	3220	code
adc-lines	10	B	adjacent_pair_states	44	state
adc-lines	10	B	visited_codes	3266	code
integrator	10	A	balance_delta	0.0021254808876186394	V
integrator	10	A	gain	-2581.6928210142723	code/V
integrator	10	A	near_balance_change	-0.2340184498996449	code
integrator	10	B	balance_delta	0.008891470489998301	V
integrator	10	B	gain	-2528.717823356551	code/V
integrator	10	B	near_balance_change	0.19092461787864753	code

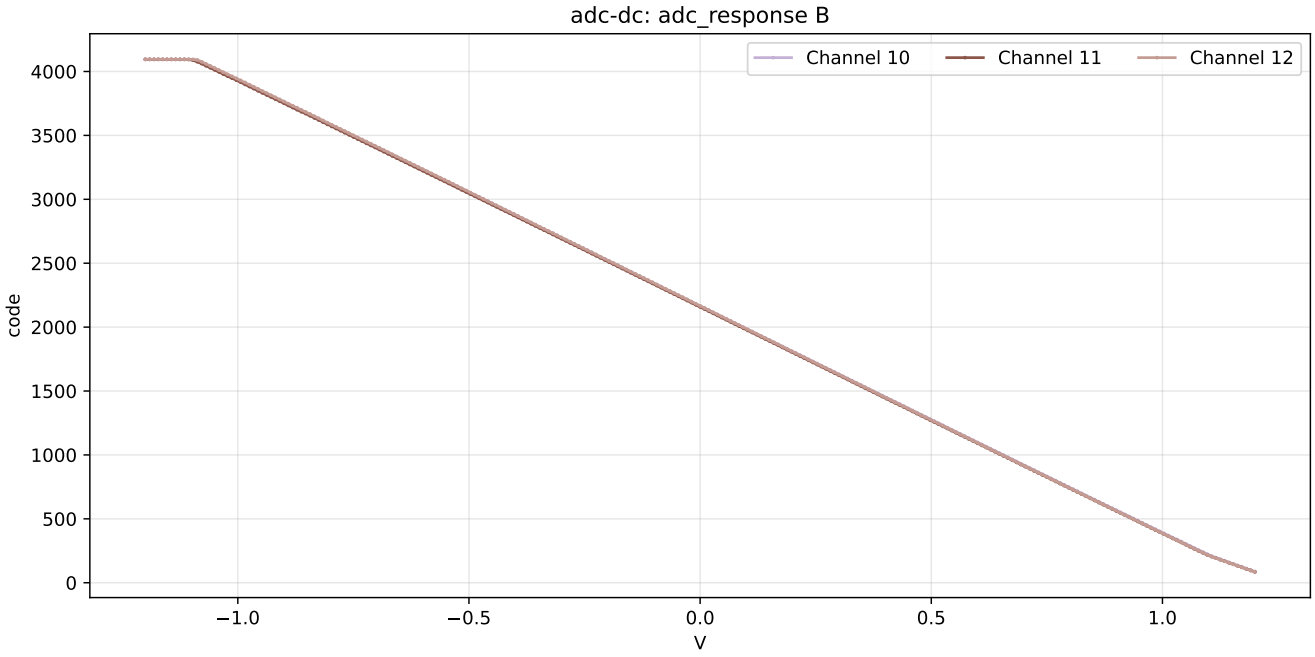
Test	Channel	Bank	Observable	Value	Unit
tdc	12		slope	-46.175872088649626	fine code/ns
tdc	12		max_fit_residual	1.7276416408801651	fine code
tdc	12		max_circular_spread	1.0733590480934954	fine code
tdc	12		events	3200	event
tdc	11		slope	-46.175079944169404	fine code/ns
tdc	11		max_fit_residual	1.6139100538513276	fine code
tdc	11		max_circular_spread	1.1070618583155902	fine code
tdc	11		events	3200	event
tdc	10		slope	-46.17427221876848	fine code/ns
tdc	10		max_fit_residual	1.1208303844473448	fine code
tdc	10		max_circular_spread	1.3046642418206191	fine code
tdc	10		events	3200	event
tdc	9		slope	-46.17113568153328	fine code/ns
tdc	9		max_fit_residual	1.5841465962344046	fine code
tdc	9		max_circular_spread	1.0614654925174123	fine code
tdc	9		events	3200	event
gate-phase	12	A	valid_points	101	point
gate-phase	12	A	observed_excursion	19.75280302671161	ns
gate-phase	12	A	end_to_end	-19.75280302671161	ns
gate-phase	12	A	largest_step	1.0396212119321913	ns
gate-phase	12	A	max_spread	0.4376432508078598	ns
gate-phase	12	B	valid_points	101	point
gate-phase	12	B	observed_excursion	19.75280302671161	ns
gate-phase	12	B	end_to_end	-19.75280302671161	ns
gate-phase	12	B	largest_step	1.0396212119321913	ns
gate-phase	12	B	max_spread	0.5191642461901361	ns
gate-phase	11	A	valid_points	101	point
gate-phase	11	A	observed_excursion	20.905062554431478	ns
gate-phase	11	A	end_to_end	-20.905062554431478	ns
gate-phase	11	A	largest_step	1.0396212119321913	ns

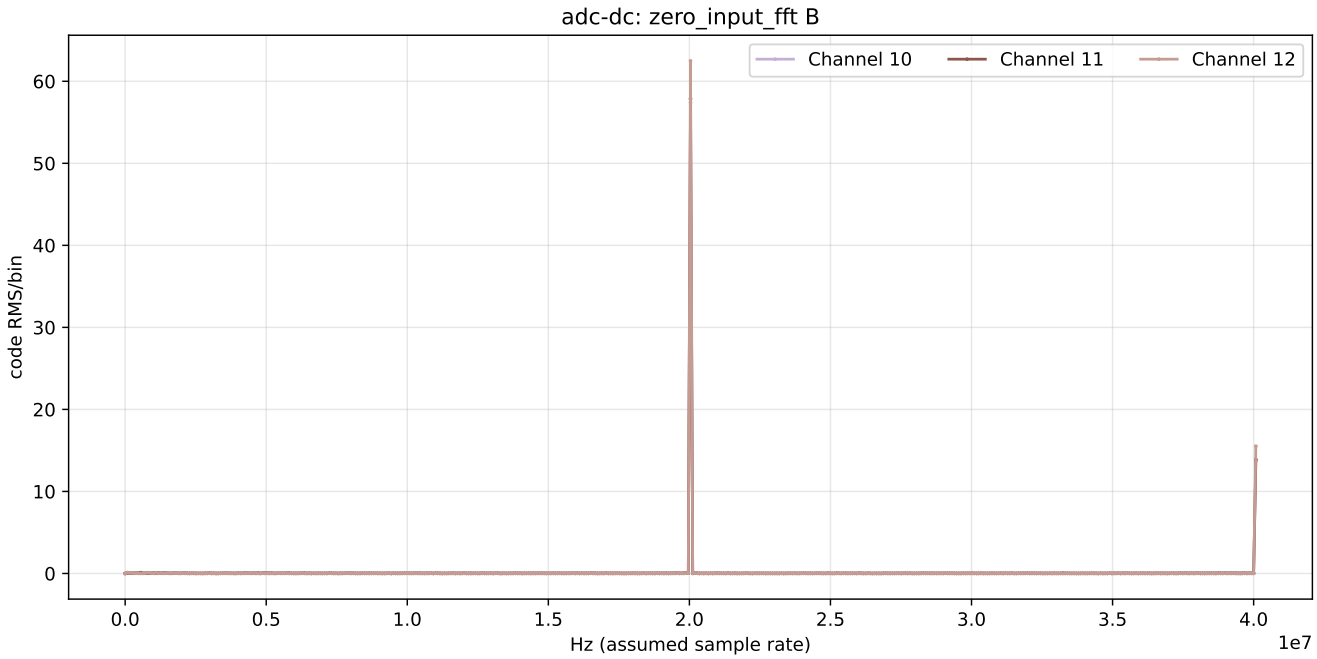
Test	Channel	Bank	Observable	Value	Unit
gate-phase	11	A	max_spread	0.517370 8034731153	ns
gate-phase	11	B	valid_points	101	point
gate-phase	11	B	observed_ excursion	20.792424 2386438	ns
gate-phase	11	B	end_to_end	-20.792424 2386438	ns
gate-phase	11	B	largest_step	1.039621 2119321913	ns
gate-phase	11	B	max_spread	0.515059 3173424455	ns
gate-phase	10	A	valid_points	101	point
gate-phase	10	A	observed_ excursion	21.832045 450575993	ns
gate-phase	10	A	end_to_end	-21.832045 450575993	ns
gate-phase	10	A	largest_step	1.039621 211932193	ns
gate-phase	10	A	max_spread	0.519701 0721490758	ns
gate-phase	10	B	valid_points	101	point
gate-phase	10	B	observed_ excursion	20.792424 2386438	ns
gate-phase	10	B	end_to_end	-20.792424 2386438	ns
gate-phase	10	B	largest_step	1.039621 2119321913	ns
gate-phase	10	B	max_spread	0.501793 7590614953	ns
gate-phase	9	A	valid_points	101	point
gate-phase	9	A	observed_ excursion	20.792424 2386438	ns
gate-phase	9	A	end_to_end	-20.792424 2386438	ns
gate-phase	9	A	largest_step	1.039621 2119321913	ns
gate-phase	9	A	max_spread	0.452989 29996023046	ns
gate-phase	9	B	valid_points	101	point
gate-phase	9	B	observed_ excursion	20.792424 2386438	ns
gate-phase	9	B	end_to_end	-20.792424 2386438	ns
gate-phase	9	B	largest_step	1.039621 2119321913	ns
gate-phase	9	B	max_spread	0.517594 5457149657	ns

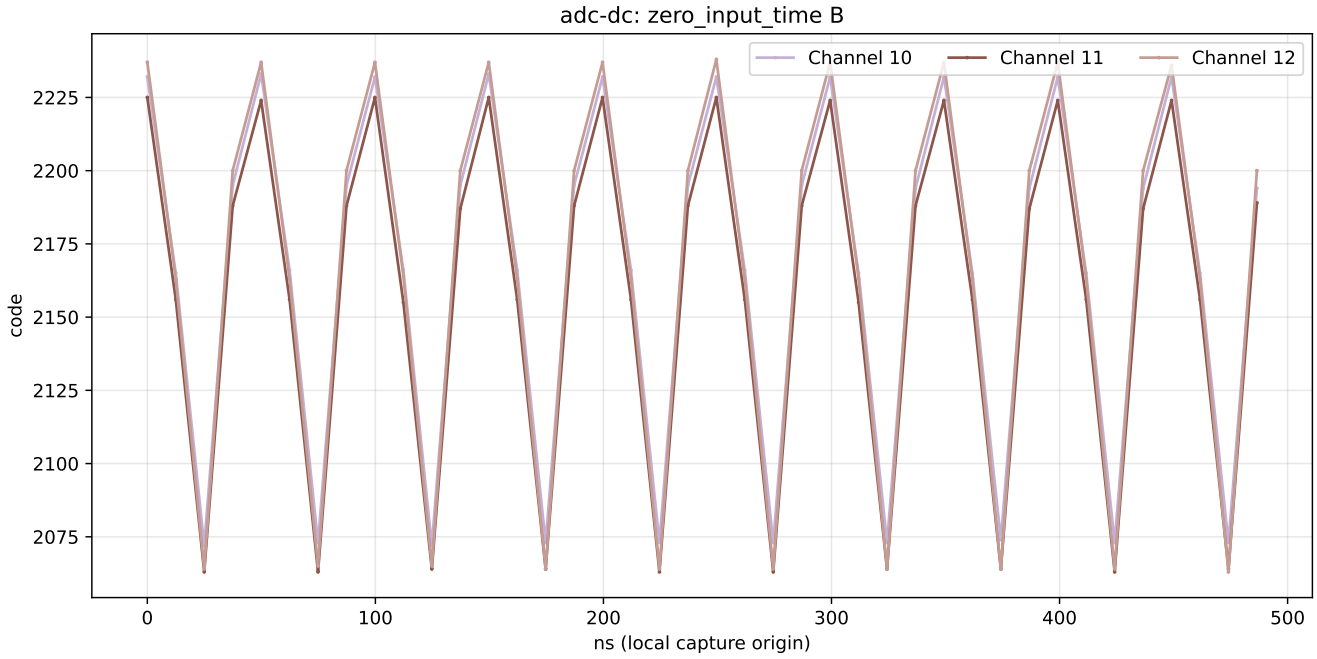

adc-dc: adc_response A
Channel 10
Channel 11
Channel 12
code
V

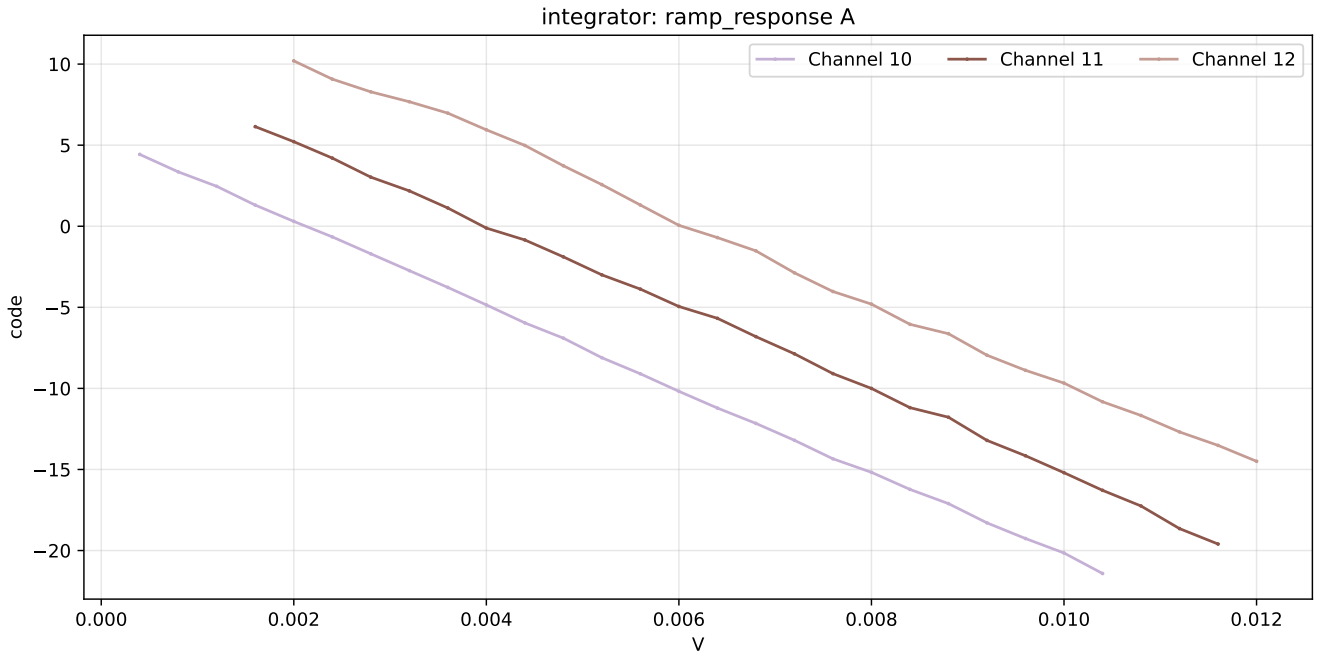

adc-dc: zero_input_fft A
Channel 10
Channel 11
Channel 12
code RMS/bin
Hz (assumed sample rate)
1e7

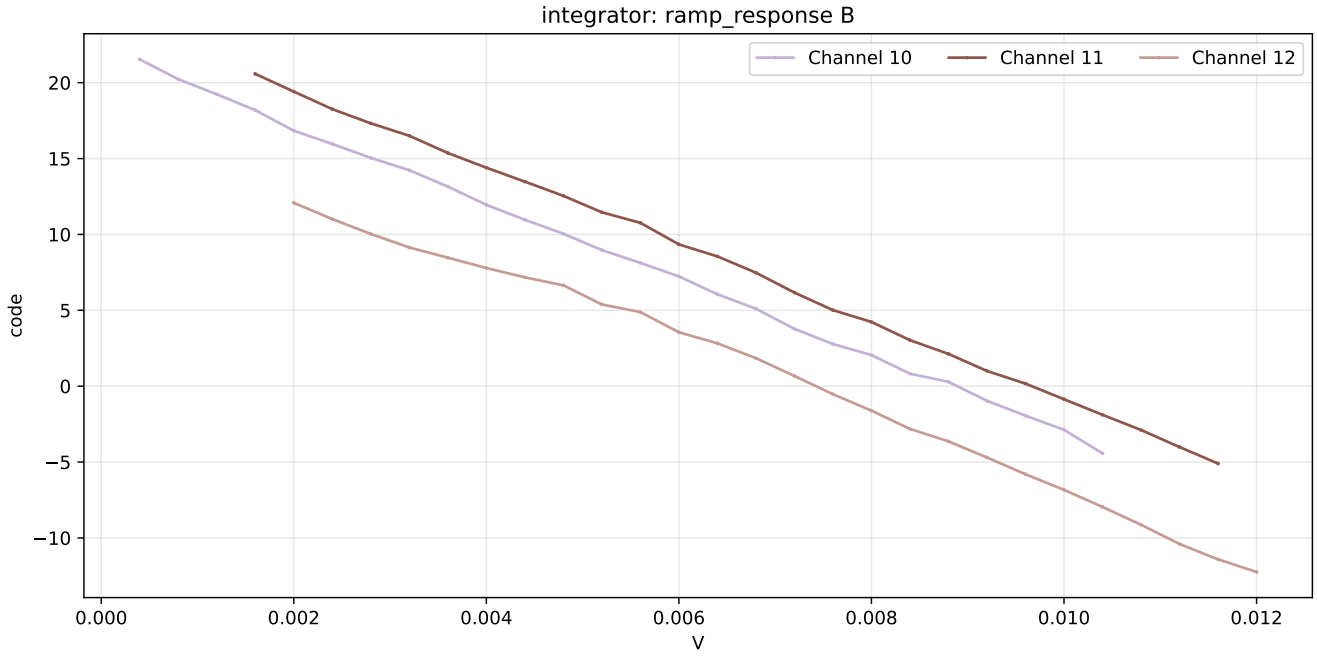

adc-dc: zero_input_time A
code
ns (local capture origin)
Channel 10
Channel 11
Channel 12
2225
2200
2175
2150
2125
2100
2075
0
100
200
300
400
500

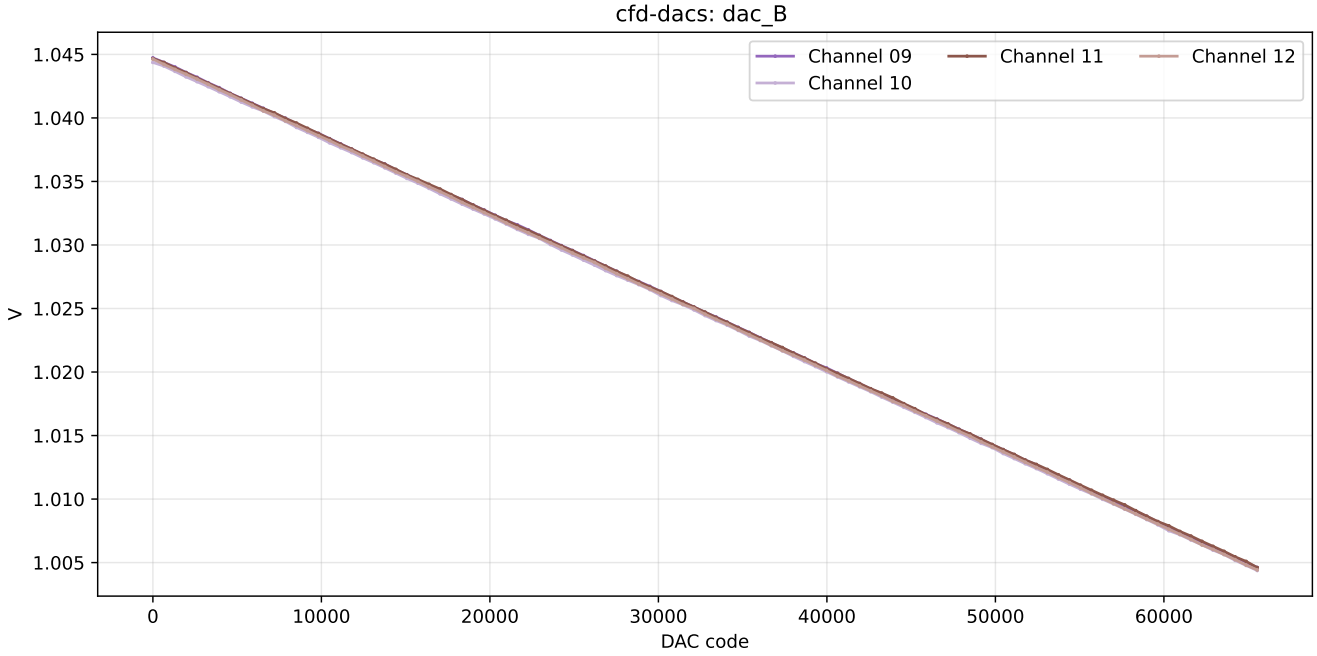

adc-dc: adc_response B
code
V
Channel 10
Channel 11
Channel 12
4000
3500
3000
2500
2000
1500
1000
500
0
−1.0
−0.5
0.0
0.5
1.0

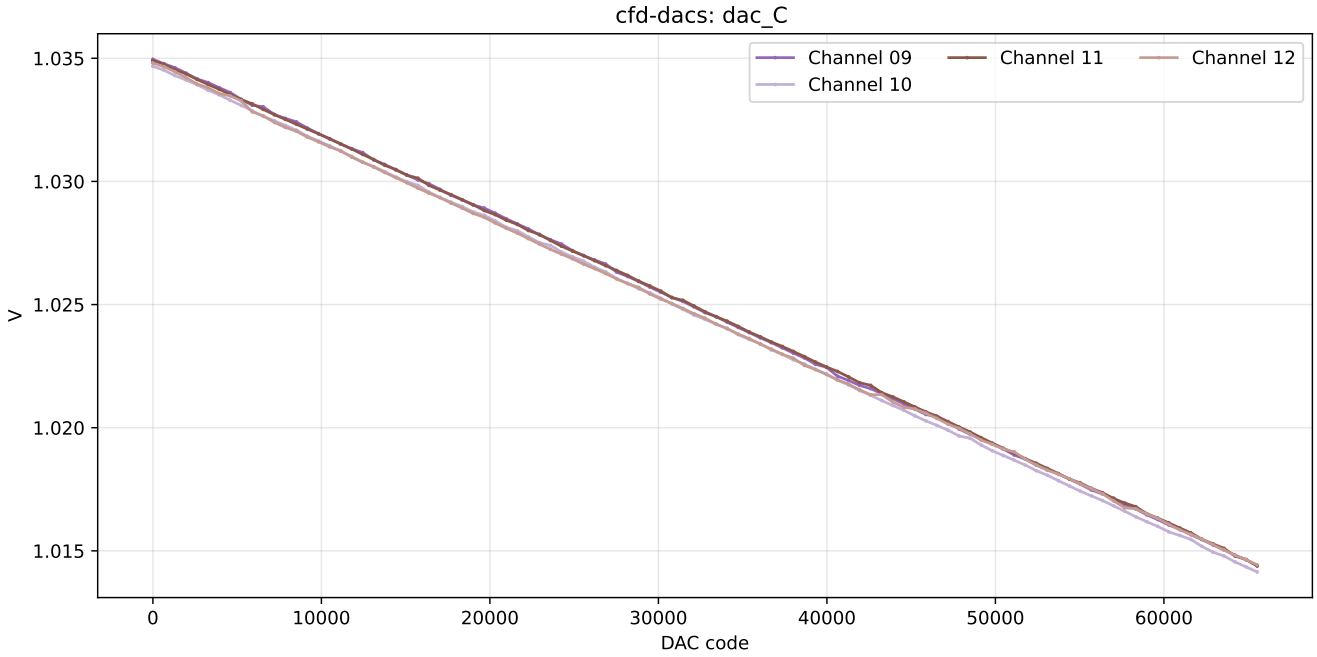

adc-dc: zero_input_fft B
Channel 10
Channel 11
Channel 12
code RMS/bin
0
10
20
30
40
50
60
0.0
0.5
1.0
1.5
2.0
2.5
3.0
3.5
4.0
1e7
Hz (assumed sample rate)

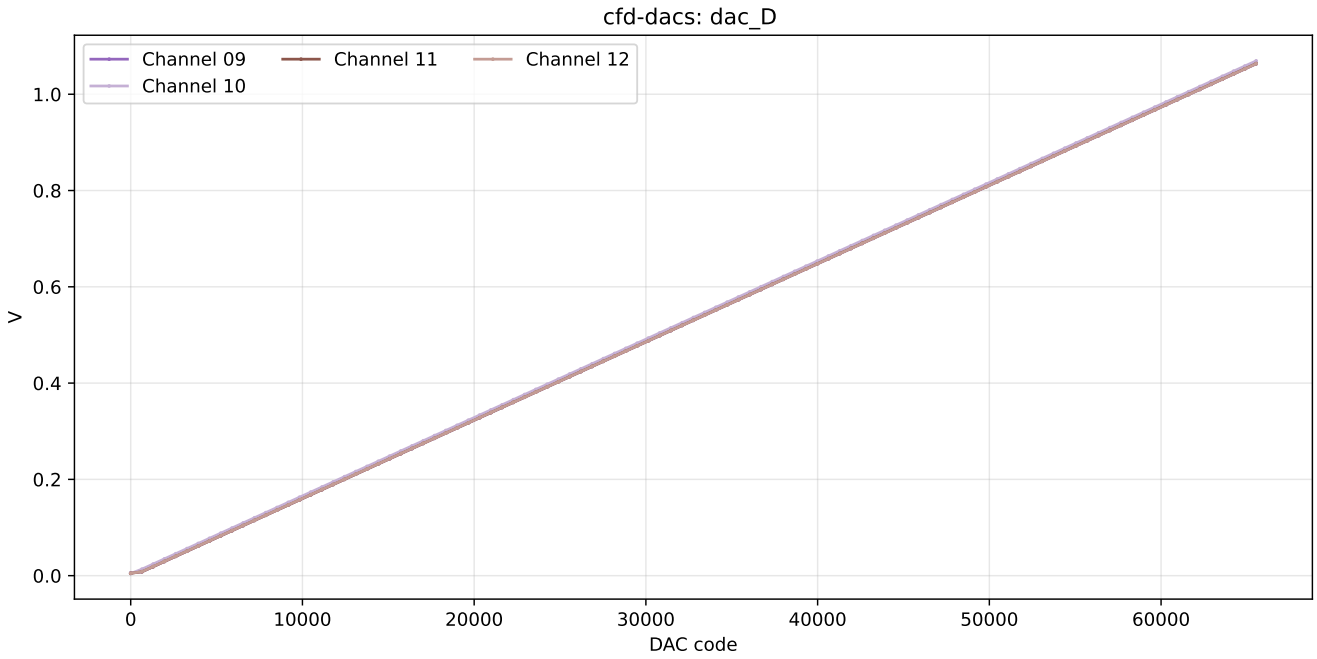

adc-dc: zero_input_time B
Channel 10
Channel 11
Channel 12
code
2075
2100
2125
2150
2175
2200
2225
0
100
200
300
400
500
ns (local capture origin)

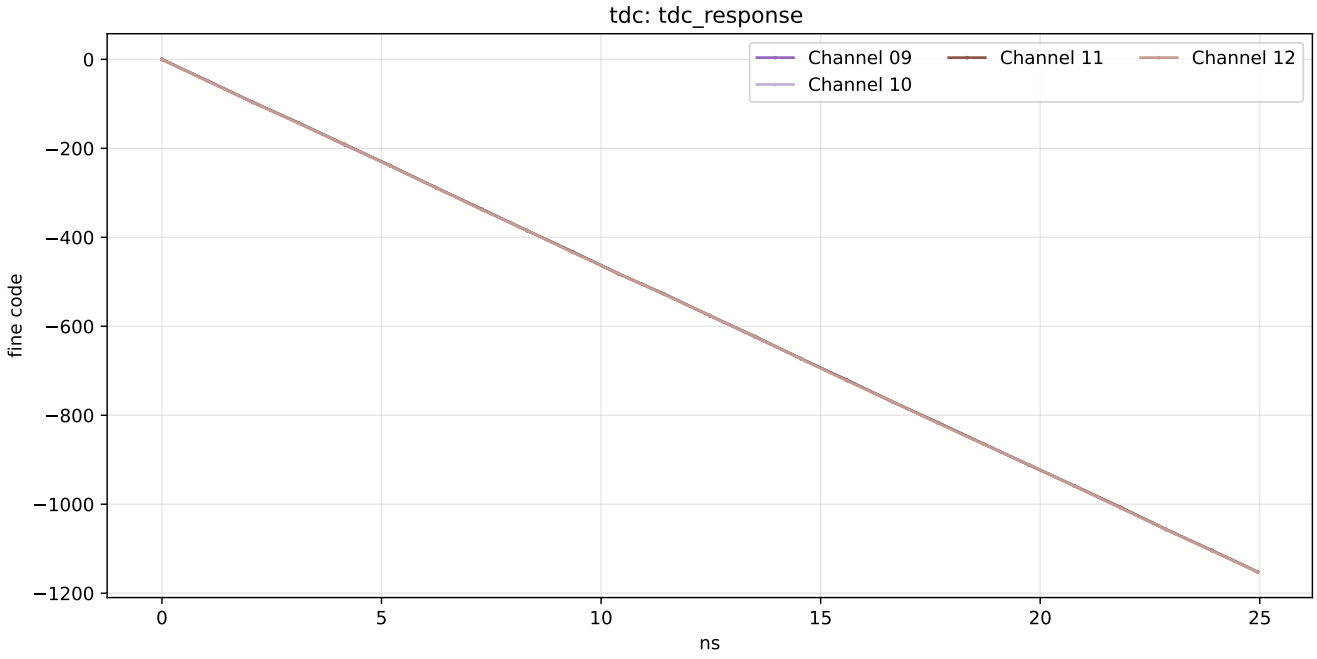

integrator: ramp_response A
Channel 10
Channel 11
Channel 12
code
V
10
5
0
−5
−10
−15
−20
0.000
0.002
0.004
0.006
0.008
0.010
0.012

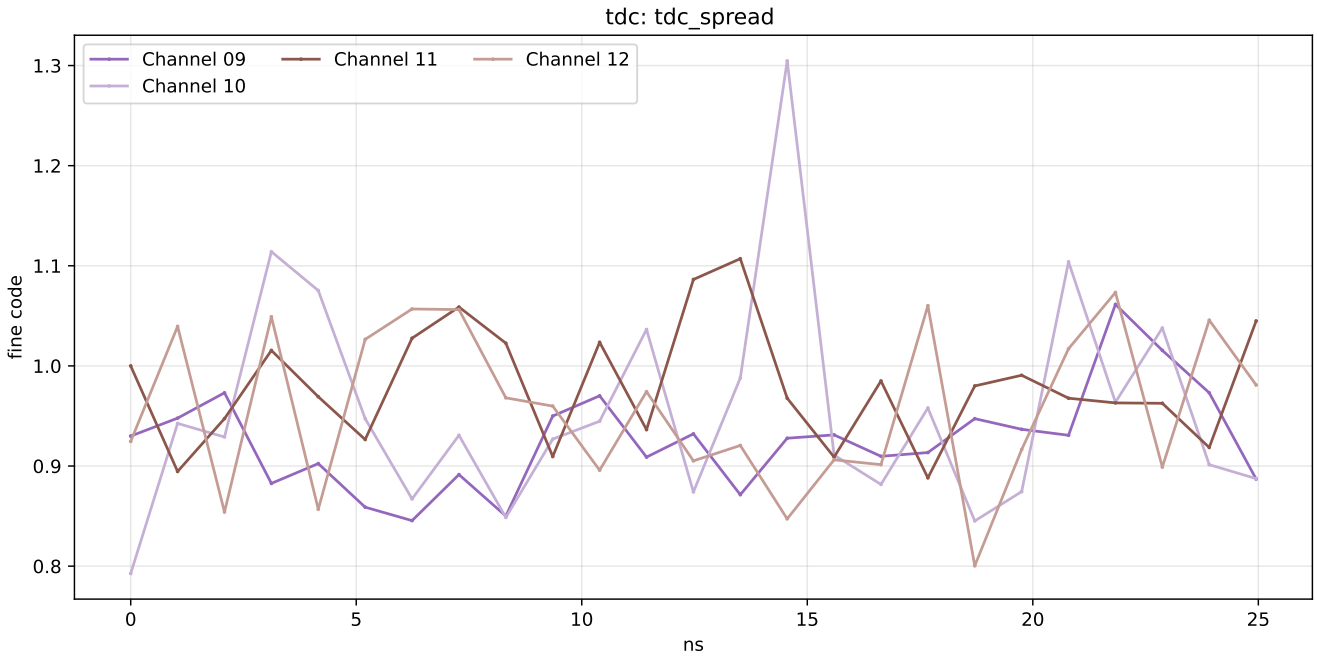

integrator: ramp_response B
Channel 10
Channel 11
Channel 12
code
V
20
15
10
5
0
−5
−10
0.000
0.002
0.004
0.006
0.008
0.010
0.012

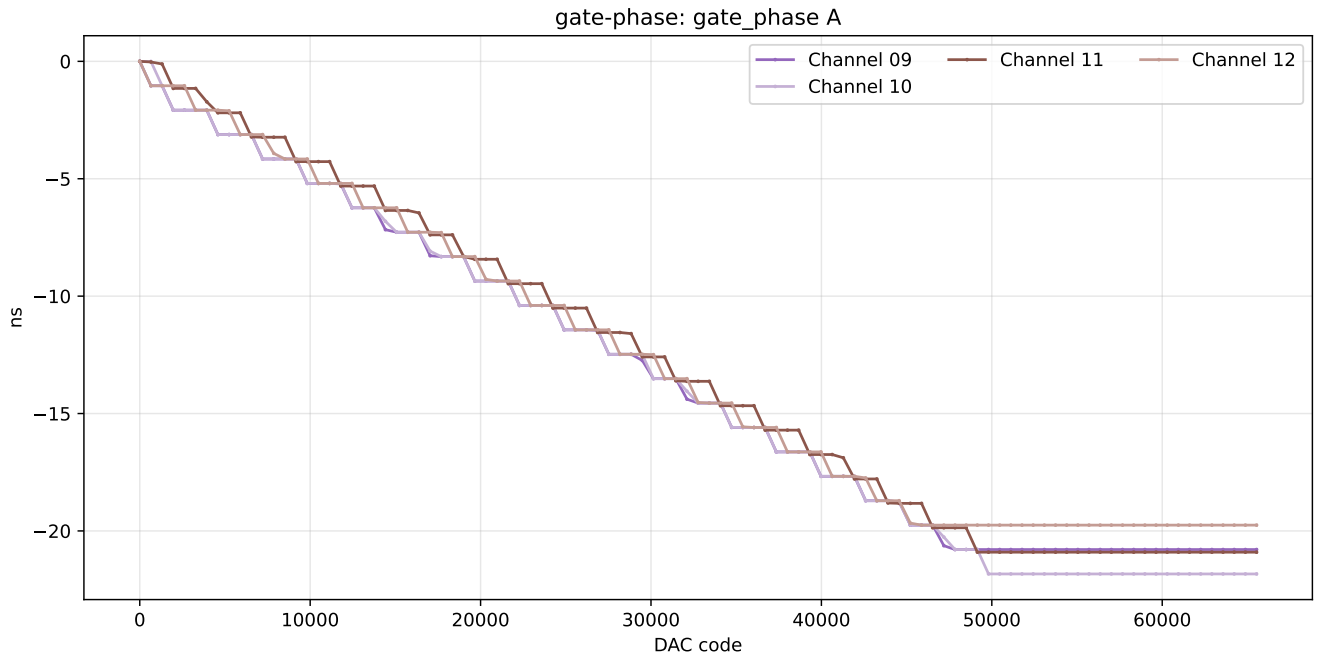

cfd-dacs: dac_B
Channel 09
Channel 10
Channel 11
Channel 12
1.045
1.040
1.035
1.030
1.025
1.020
1.015
1.010
1.005
V
0
10000
20000
30000
40000
50000
60000
DAC code

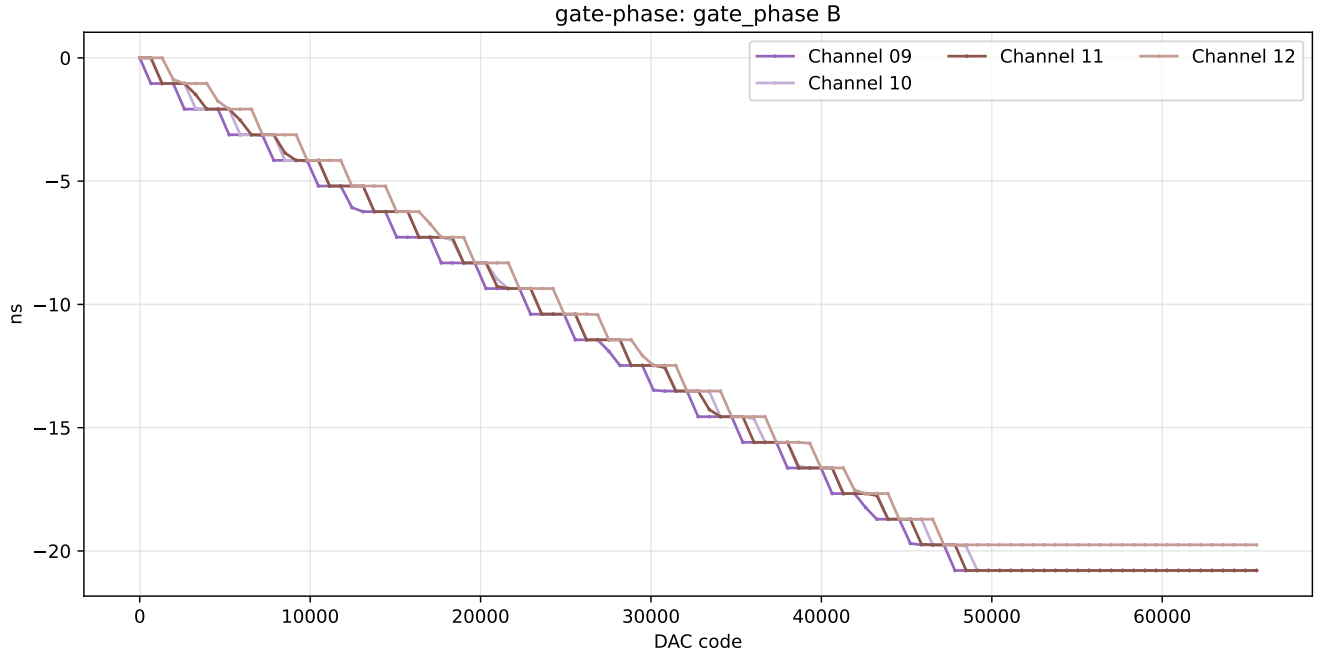

cfd-dacs: dac_C
Channel 09
Channel 10
Channel 11
Channel 12
1.035
1.030
1.025
1.020
1.015
V
0
10000
20000
30000
40000
50000
60000
DAC code

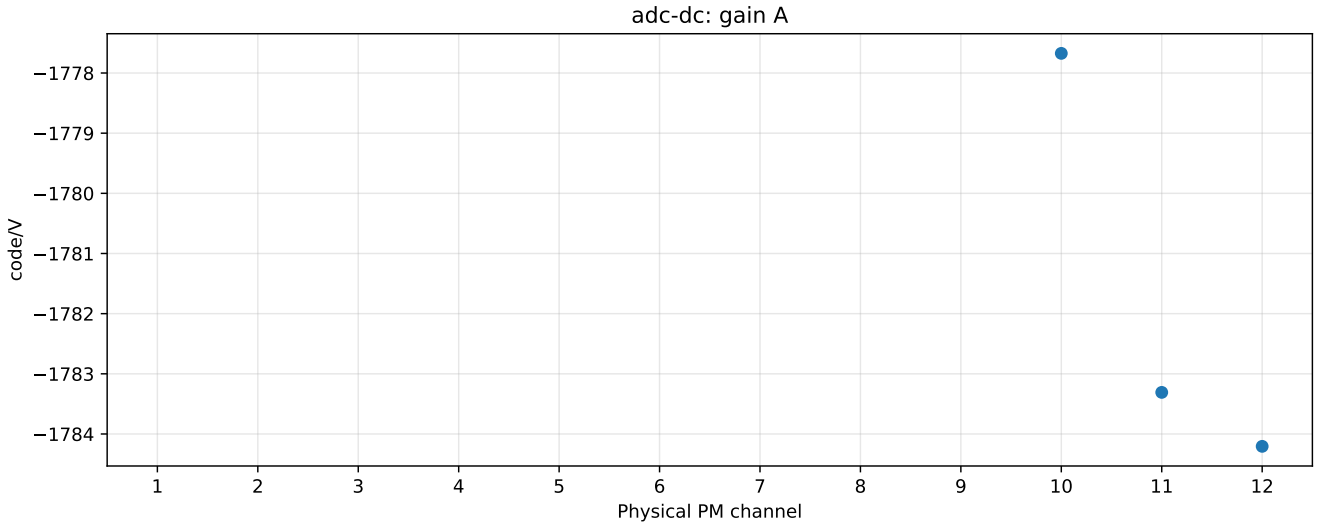

cfd-dacs: dac_D
Channel 09
Channel 11
Channel 12
Channel 10
V
DAC code

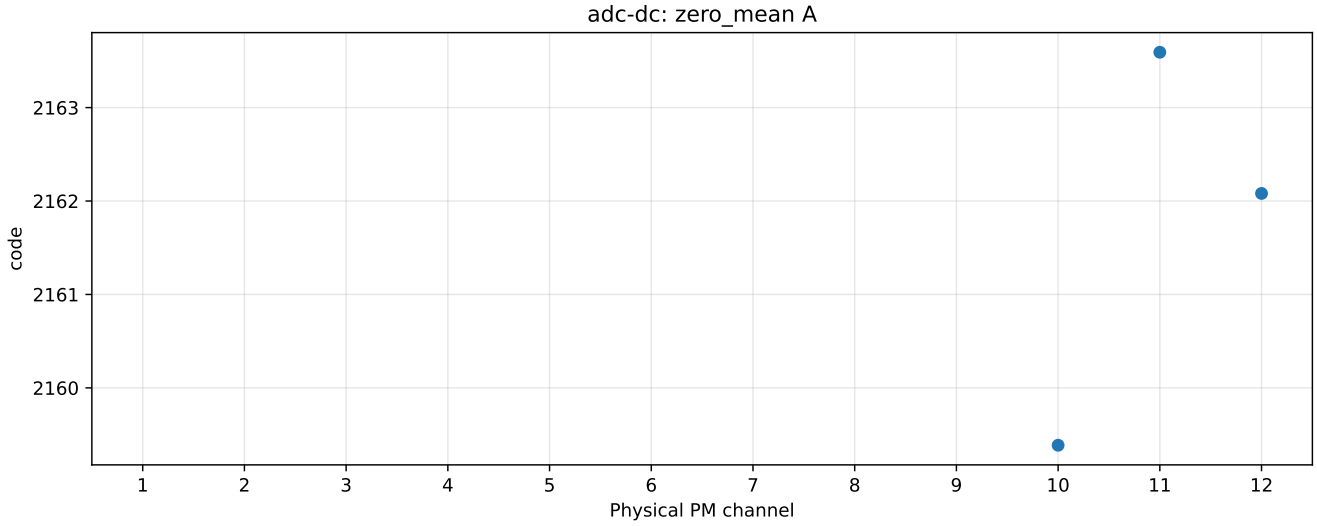

tdc: tdc_response
Channel 09
Channel 11
Channel 12
Channel 10
fine code
ns

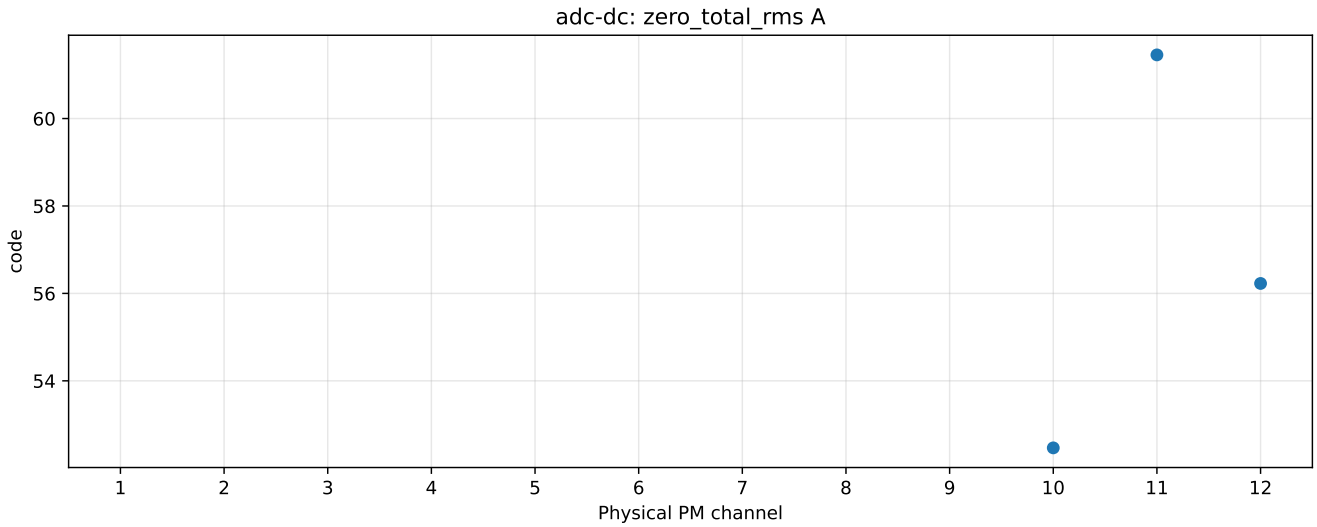

tdc: tdc_spread
Channel 09
Channel 10
Channel 11
Channel 12
fine code
1.3
1.2
1.1
1.0
0.9
0.8
0
5
10
15
20
25
ns

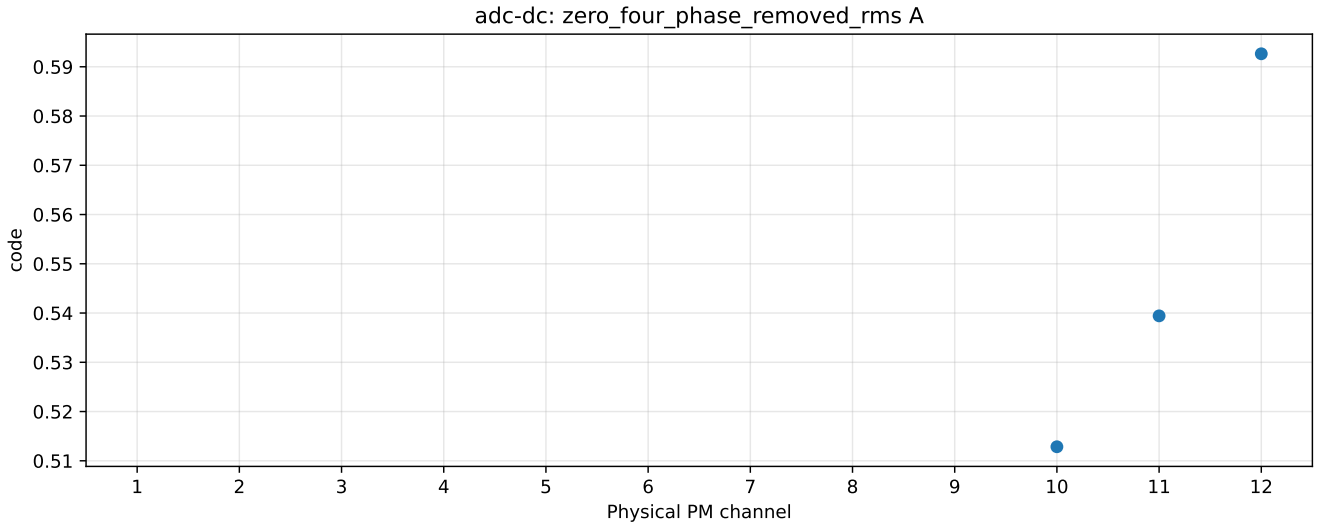

gate-phase: gate_phase A
Channel 09
Channel 10
Channel 11
Channel 12
ns
0
−5
−10
−15
−20
0
10000
20000
30000
40000
50000
60000
DAC code

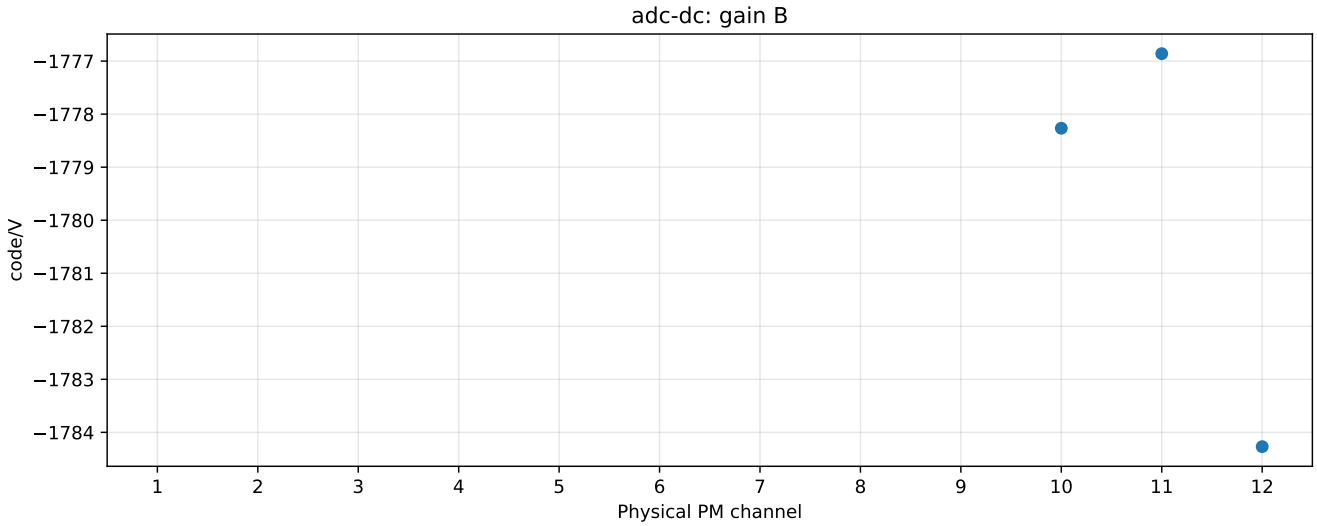

gate-phase: gate_phase B
Channel 09
Channel 10
Channel 11
Channel 12
ns
0
−5
−10
−15
−20
0
10000
20000
30000
40000
50000
60000
DAC code

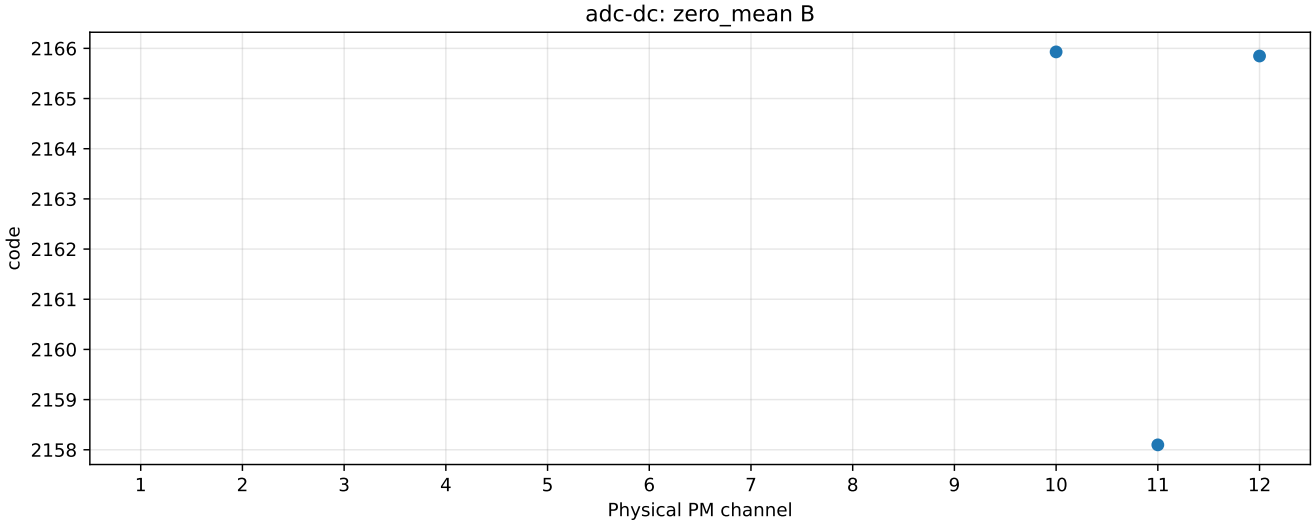

adc-dc: gain A
code/V
−1778
−1779
−1780
−1781
−1782
−1783
−1784
1
2
3
4
5
6
7
8
9
10
11
12
Physical PM channel

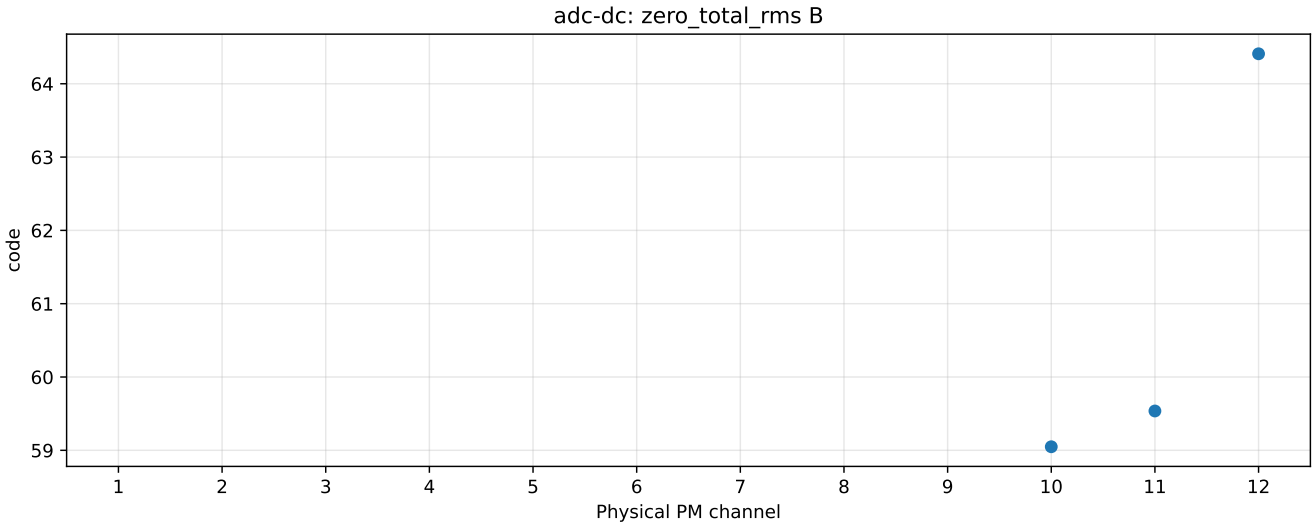

adc-dc: zero_mean A
code
2163
2162
2161
2160
1
2
3
4
5
6
7
8
9
10
11
12
Physical PM channel

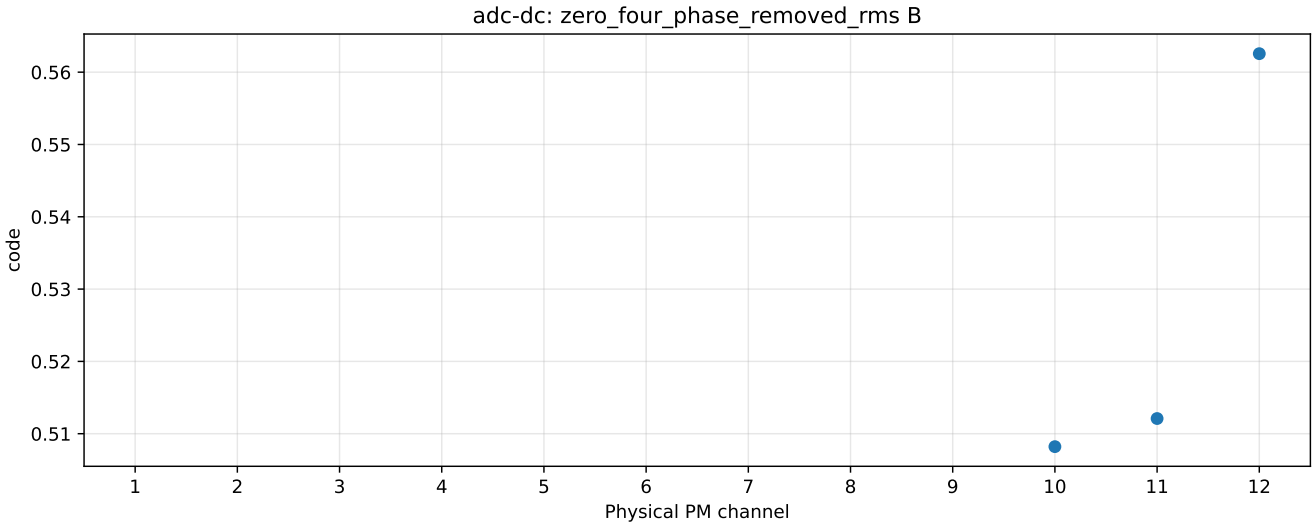

adc-dc: zero_total_rms A
code
60
58
56
54
1
2
3
4
5
6
7
8
9
10
11
12
Physical PM channel

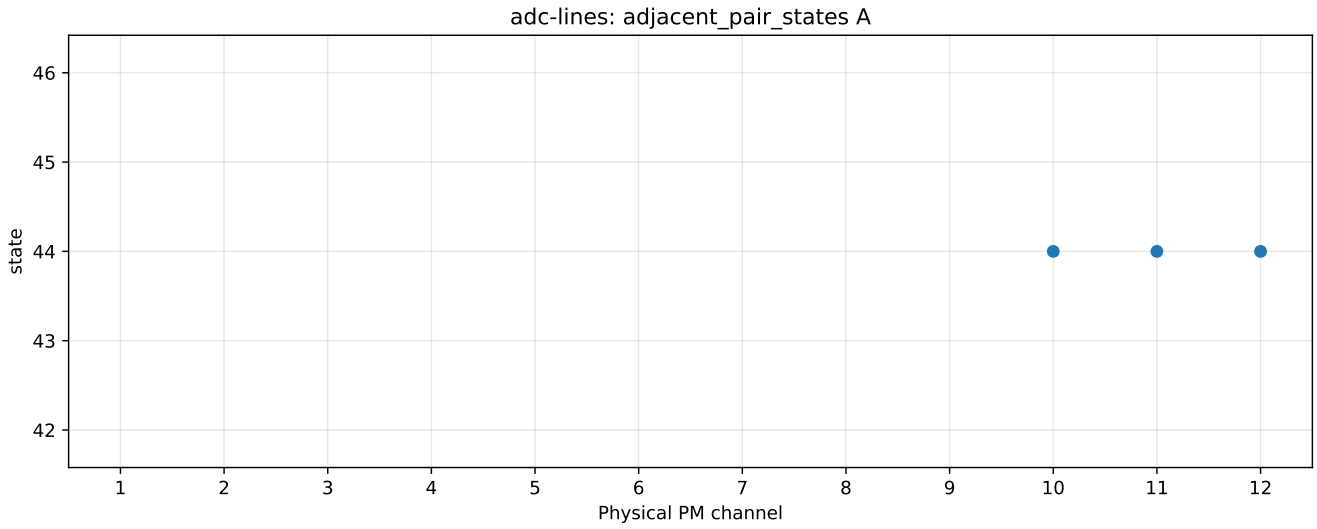

adc-dc: zero_four_phase_removed_rms A
code
0.59
0.58
0.57
0.56
0.55
0.54
0.53
0.52
0.51
1
2
3
4
5
6
7
8
9
10
11
12
Physical PM channel

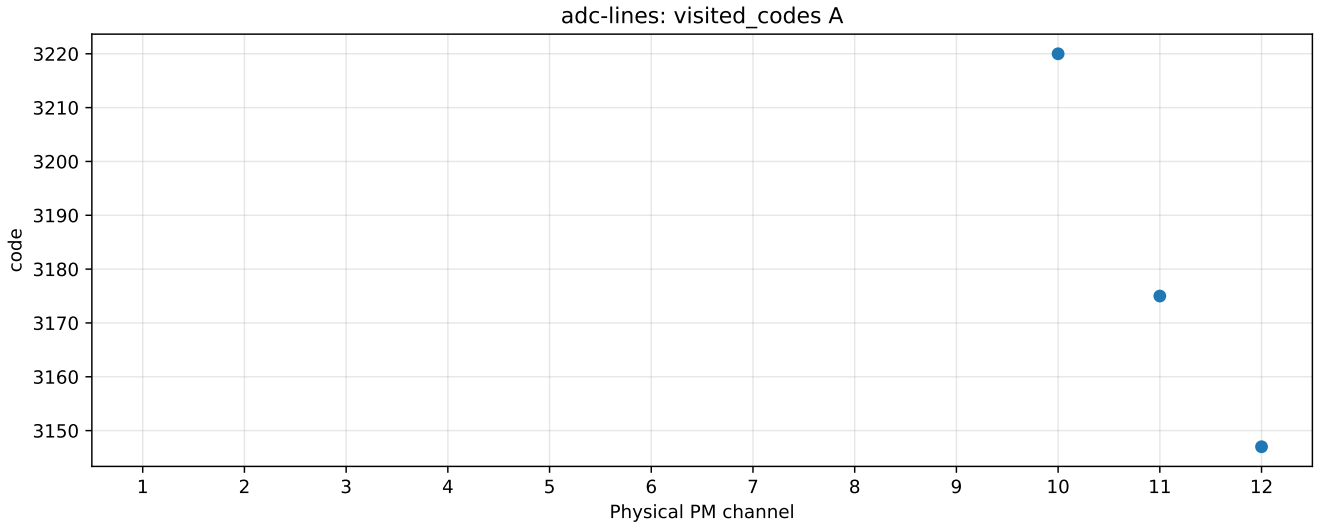

adc-dc: gain B
code/V
−1777
−1778
−1779
−1780
−1781
−1782
−1783
−1784
1
2
3
4
5
6
7
8
9
10
11
12
Physical PM channel

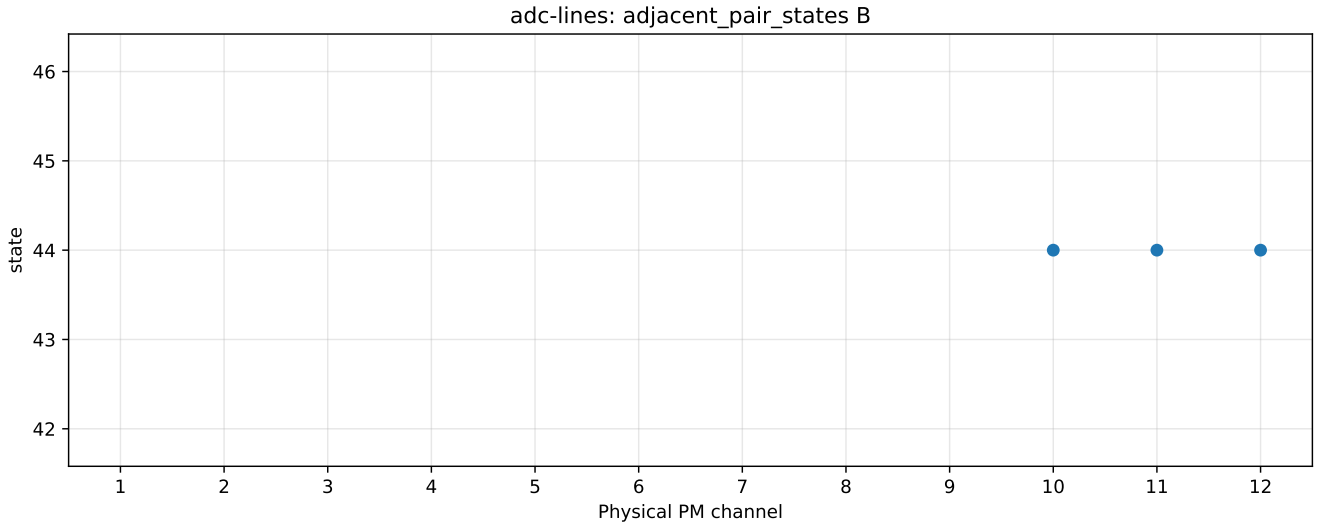

adc-dc: zero_mean B
code
2166
2165
2164
2163
2162
2161
2160
2159
2158
1
2
3
4
5
6
7
8
9
10
11
12
Physical PM channel

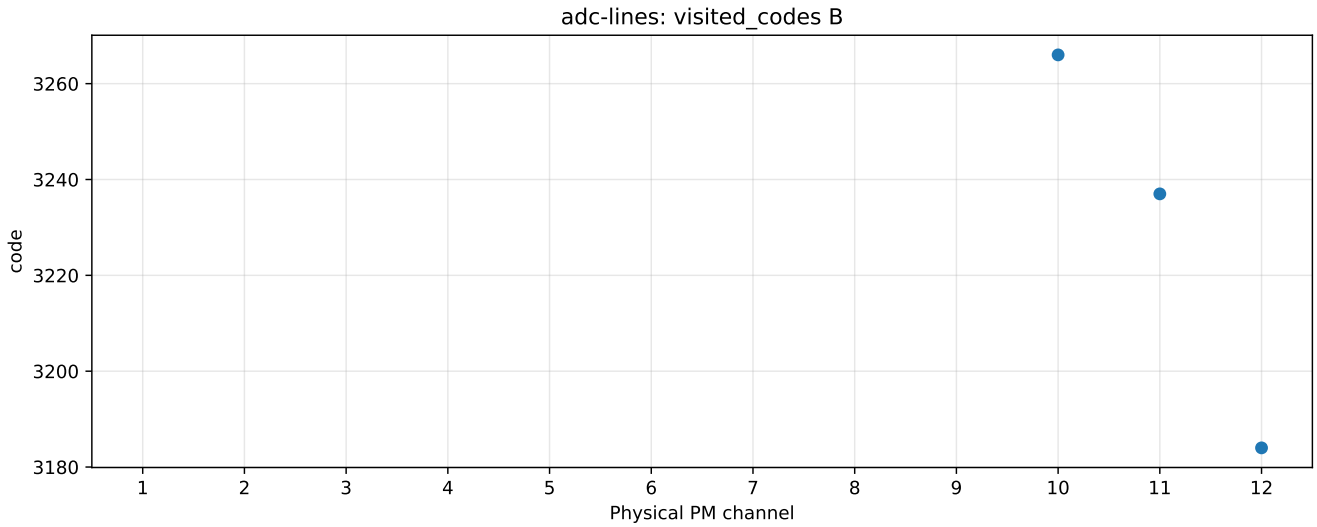

adc-dc: zero_total_rms B
code
64
63
62
61
60
59
1
2
3
4
5
6
7
8
9
10
11
12
Physical PM channel

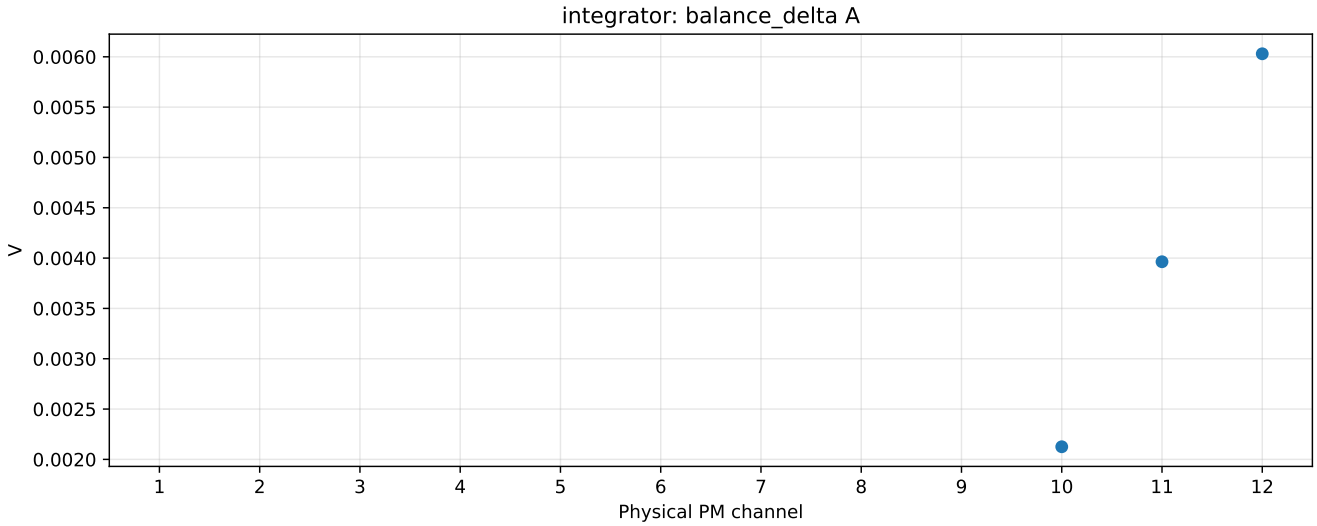

adc-dc: zero_four_phase_removed_rms B
code
0.56
0.55
0.54
0.53
0.52
0.51
1
2
3
4
5
6
7
8
9
10
11
12
Physical PM channel

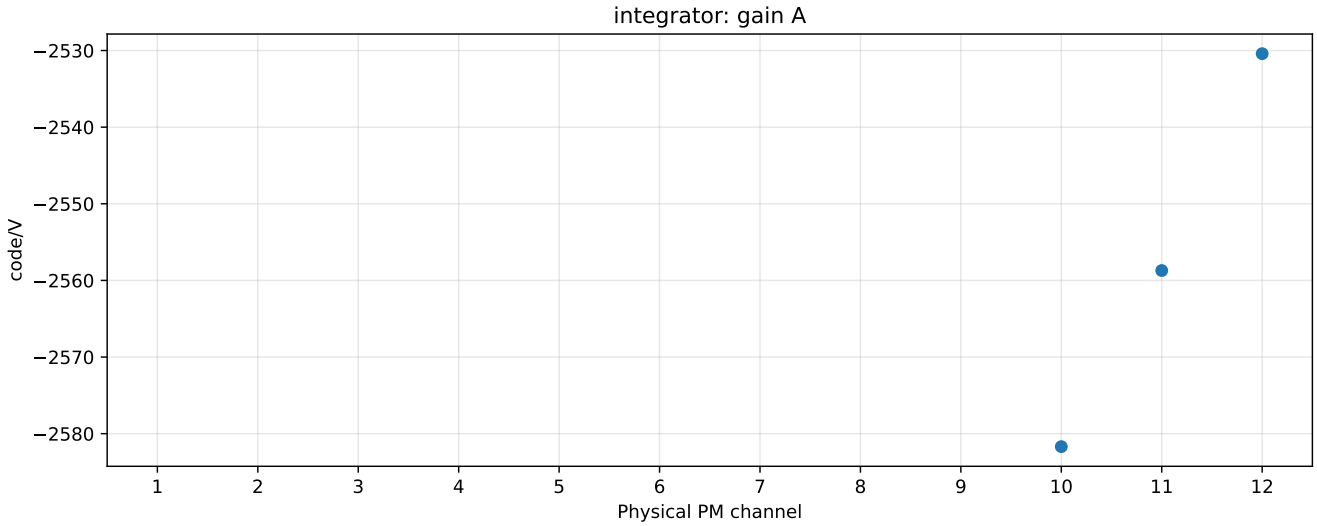

adc-lines: adjacent_pair_states A
state
46
45
44
43
42
1
2
3
4
5
6
7
8
9
10
11
12
Physical PM channel

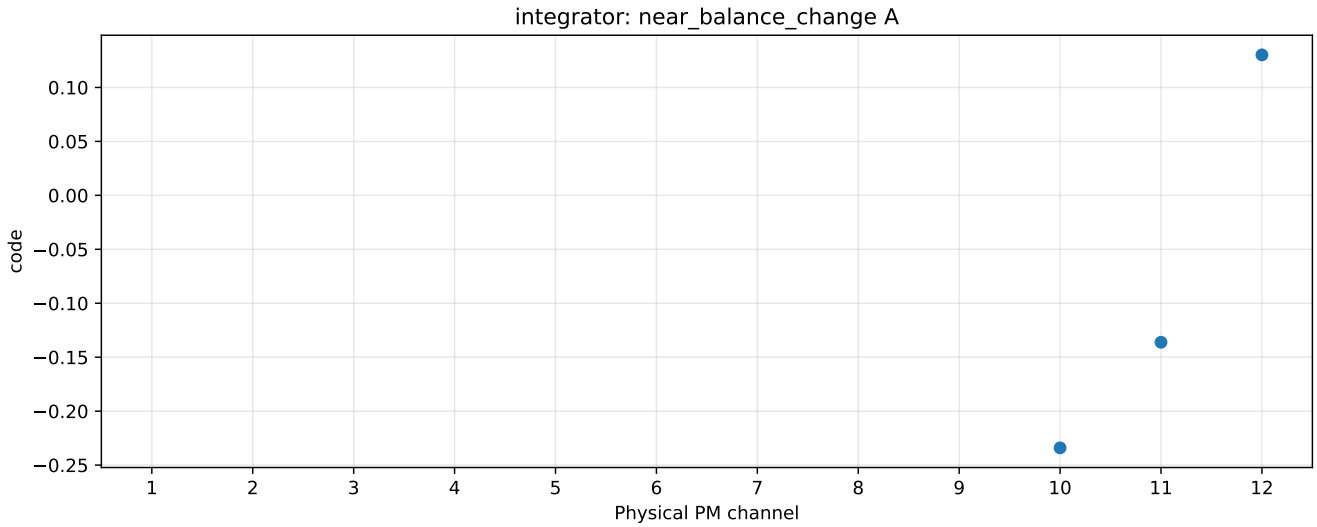

adc-lines: visited_codes A
code
3220
3210
3200
3190
3180
3170
3160
3150
1
2
3
4
5
6
7
8
9
10
11
12
Physical PM channel

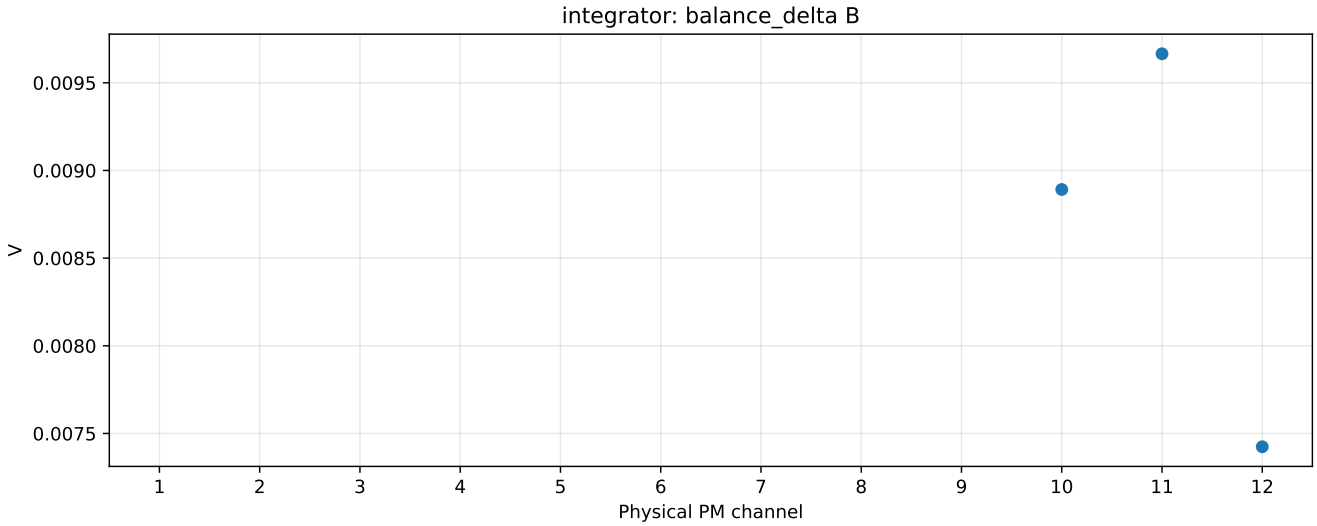

adc-lines: adjacent_pair_states B
state
46
45
44
43
42
1
2
3
4
5
6
7
8
9
10
11
12
Physical PM channel

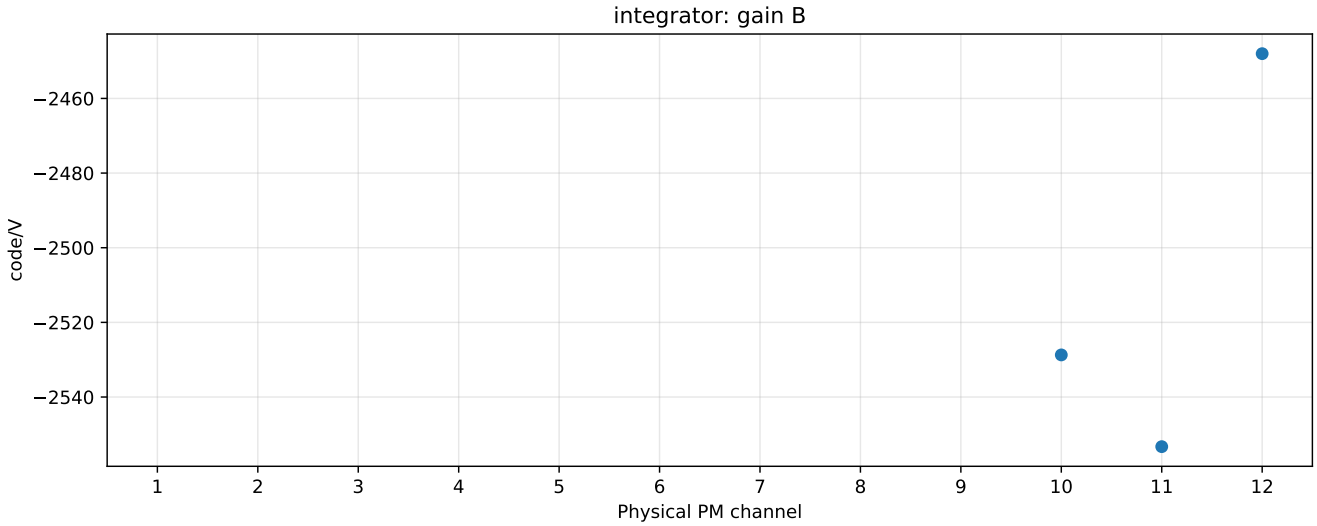

adc-lines: visited_codes B
code
3260
3240
3220
3200
3180
1
2
3
4
5
6
7
8
9
10
11
12
Physical PM channel

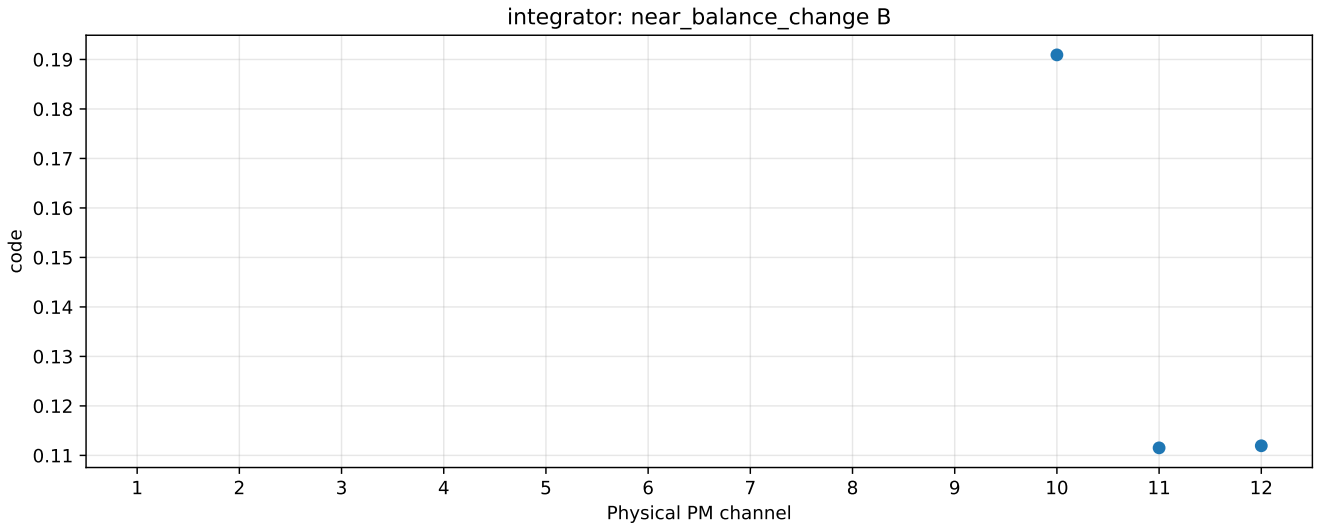

integrator: balance_delta A
V
0.0060
0.0055
0.0050
0.0045
0.0040
0.0035
0.0030
0.0025
0.0020
1
2
3
4
5
6
7
8
9
10
11
12
Physical PM channel

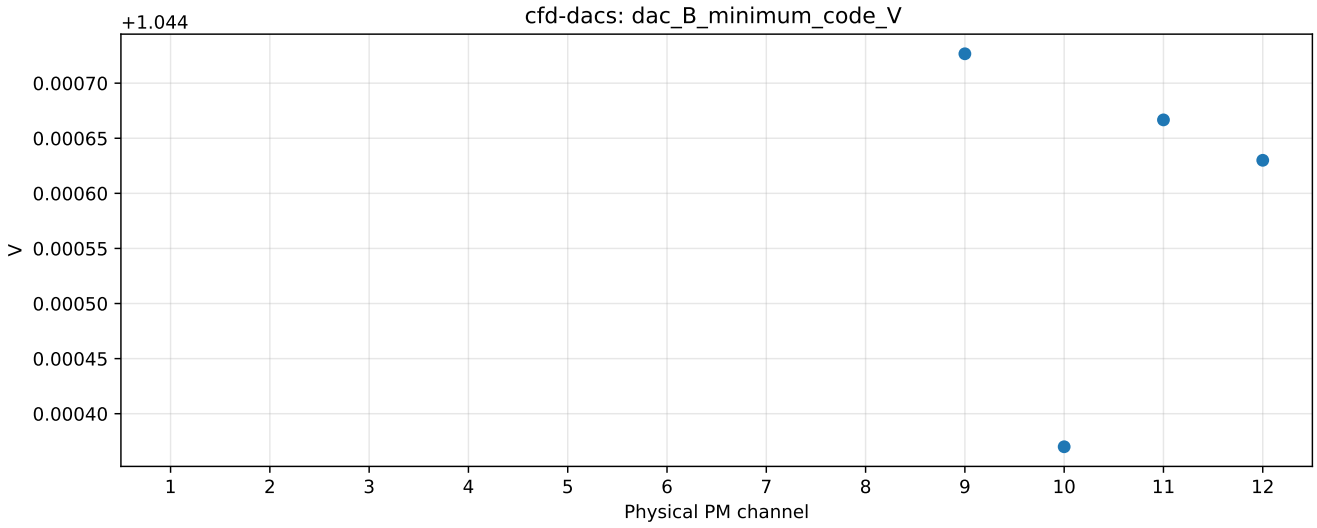

integrator: gain A
code/V
−2530
−2540
−2550
−2560
−2570
−2580
1
2
3
4
5
6
7
8
9
10
11
12
Physical PM channel

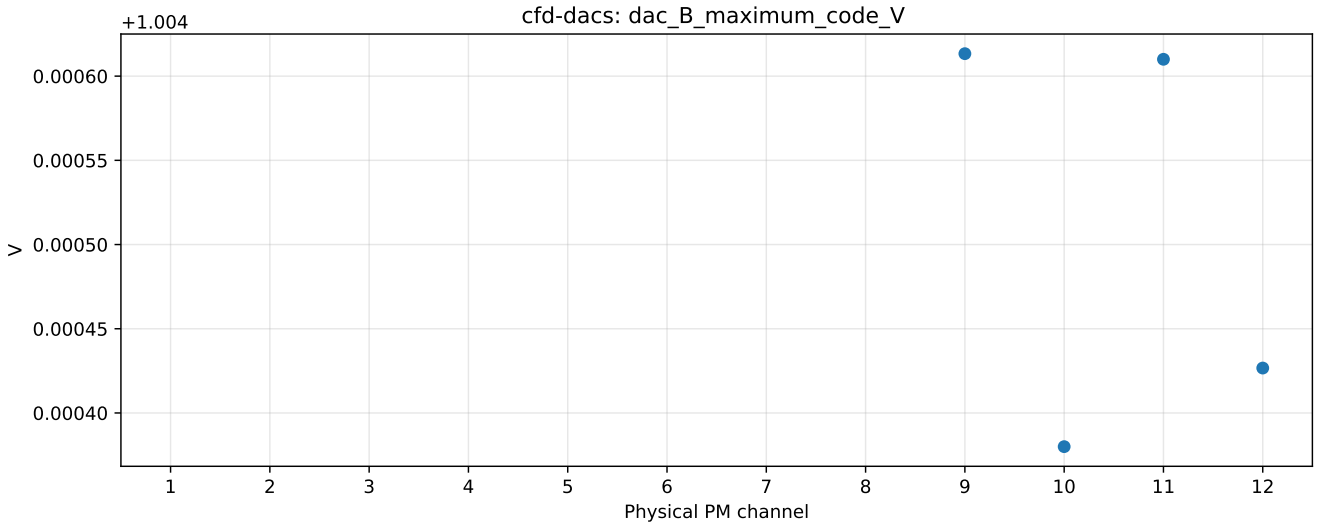

integrator: near_balance_change A
code
0.10
0.05
0.00
−0.05
−0.10
−0.15
−0.20
−0.25
1
2
3
4
5
6
7
8
9
10
11
12
Physical PM channel

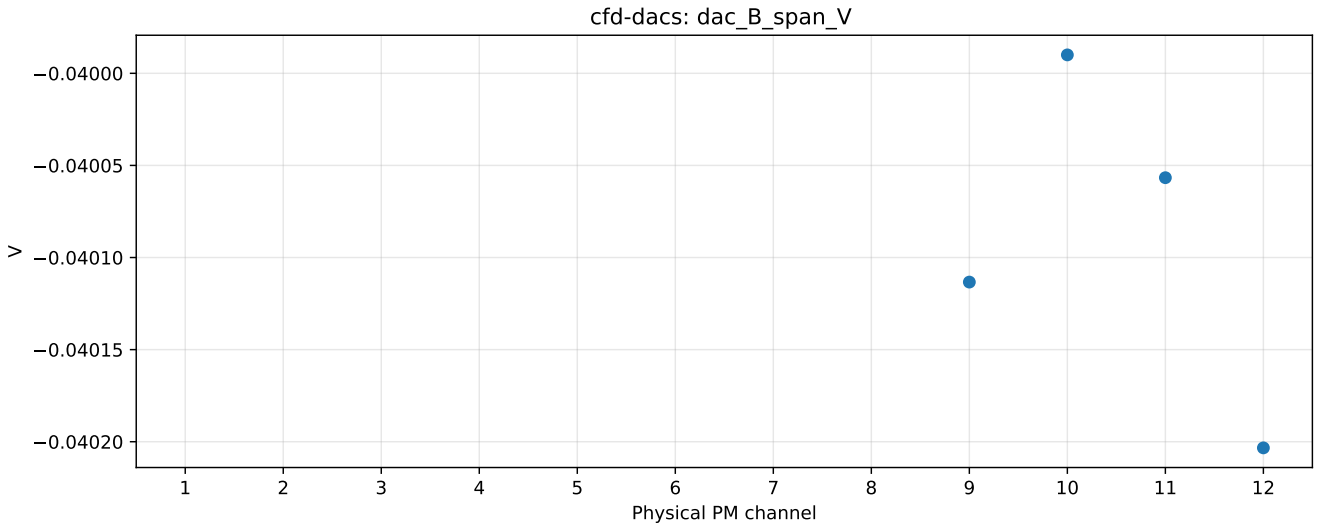

integrator: balance_delta B
V
0.0095
0.0090
0.0085
0.0080
0.0075
1
2
3
4
5
6
7
8
9
10
11
12
Physical PM channel

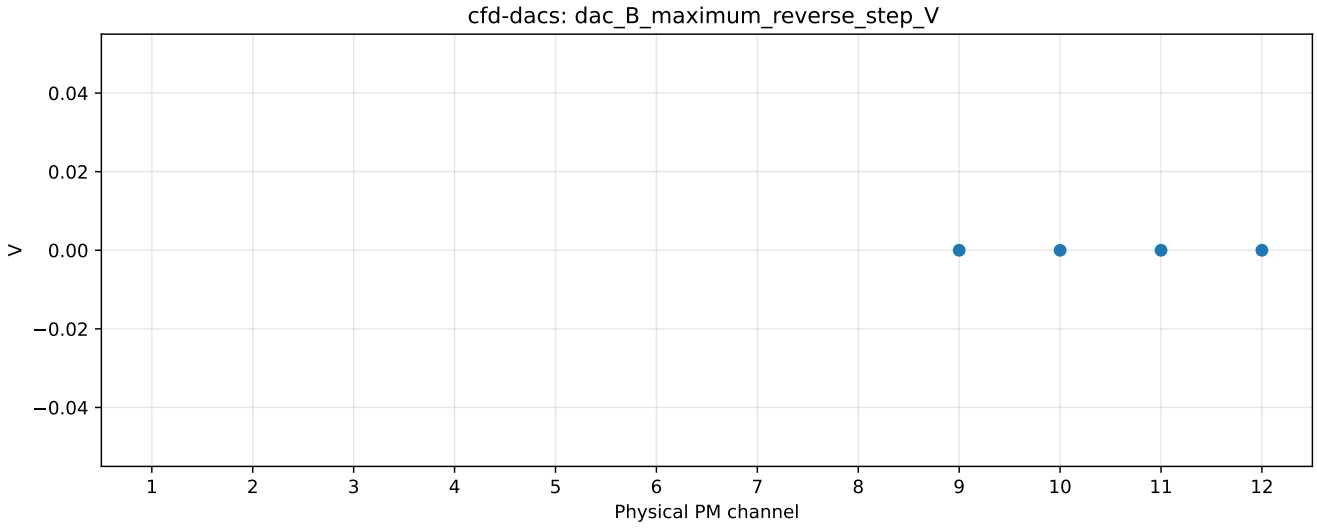

integrator: gain B
code/V
−2460
−2480
−2500
−2520
−2540
1
2
3
4
5
6
7
8
9
10
11
12
Physical PM channel

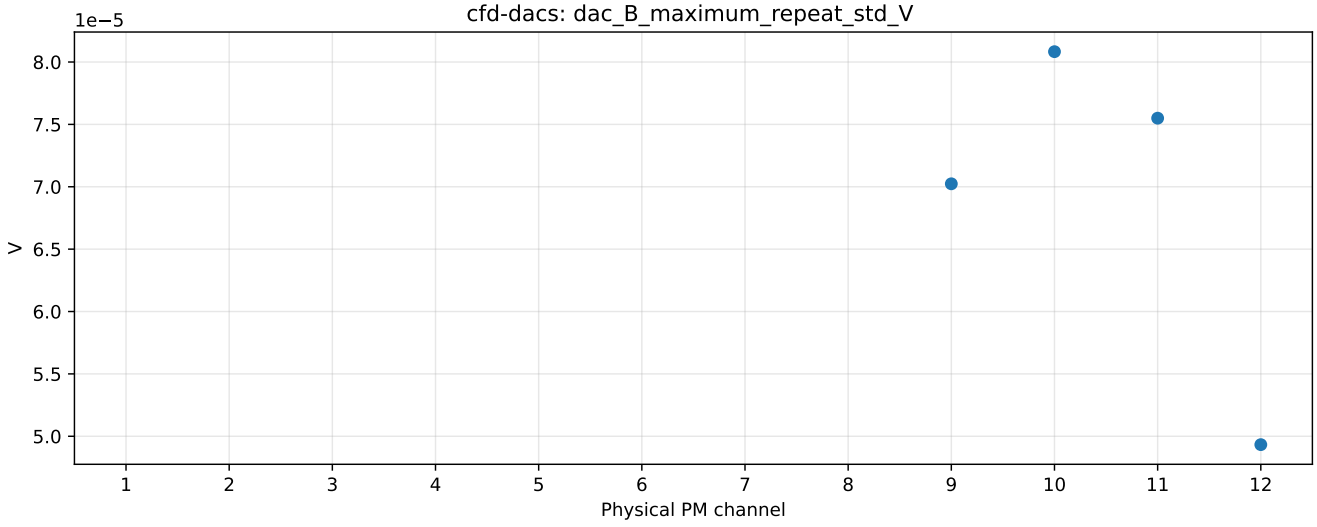

integrator: near_balance_change B
code
0.19
0.18
0.17
0.16
0.15
0.14
0.13
0.12
0.11
1
2
3
4
5
6
7
8
9
10
11
12
Physical PM channel

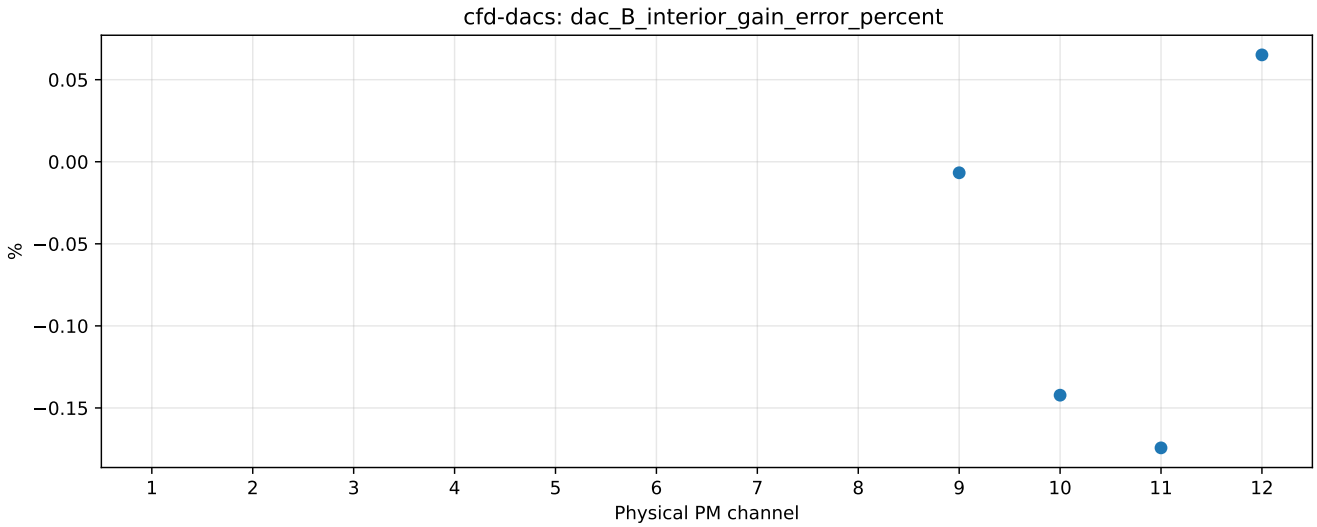

cfd-dacs: dac_B_minimum_code_V
+1.044
V
0.00070
0.00065
0.00060
0.00055
0.00050
0.00045
0.00040
1
2
3
4
5
6
7
8
9
10
11
12
Physical PM channel

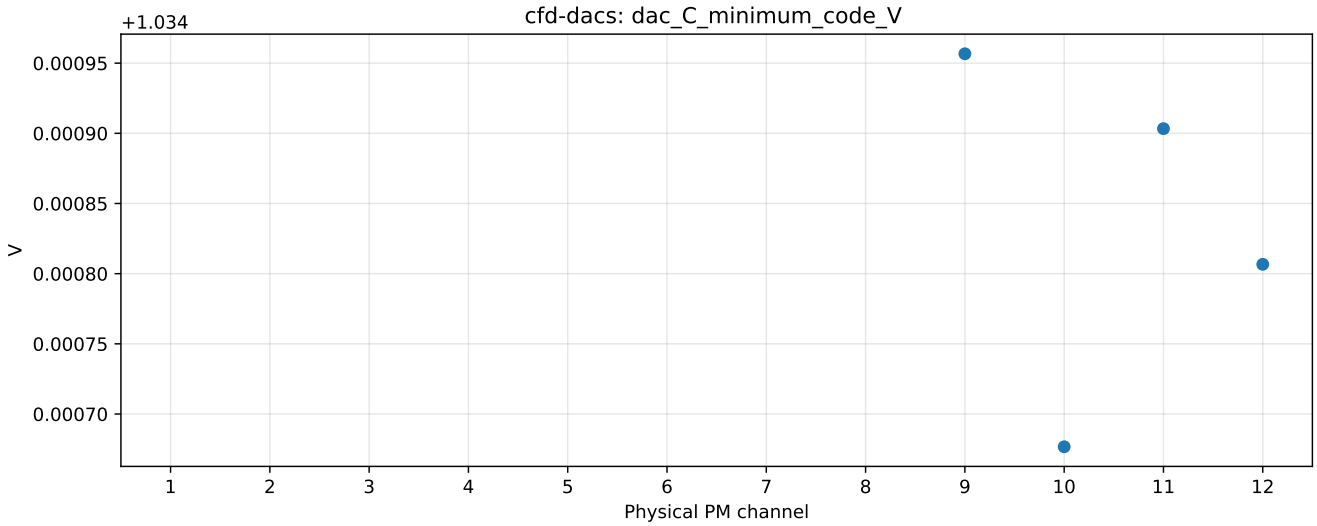

cfd-dacs: dac_B_maximum_code_V
+1.004
V
0.00060
0.00055
0.00050
0.00045
0.00040
1
2
3
4
5
6
7
8
9
10
11
12
Physical PM channel

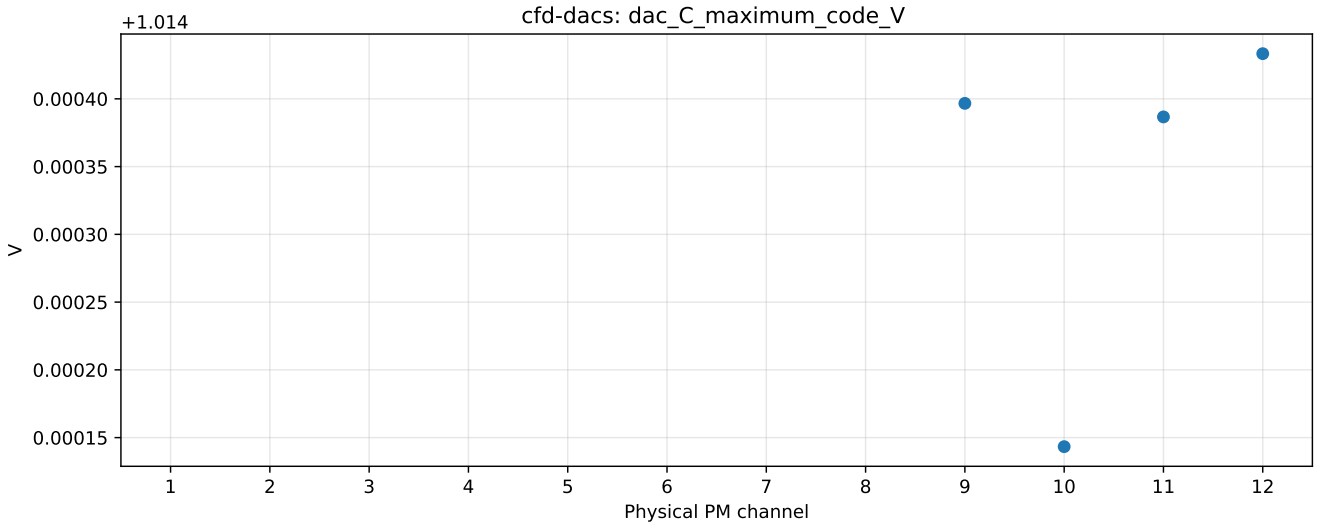

cfd-dacs: dac_B_span_V
−0.04000
−0.04005
−0.04010
−0.04015
−0.04020
V
1
2
3
4
5
6
7
8
9
10
11
12
Physical PM channel

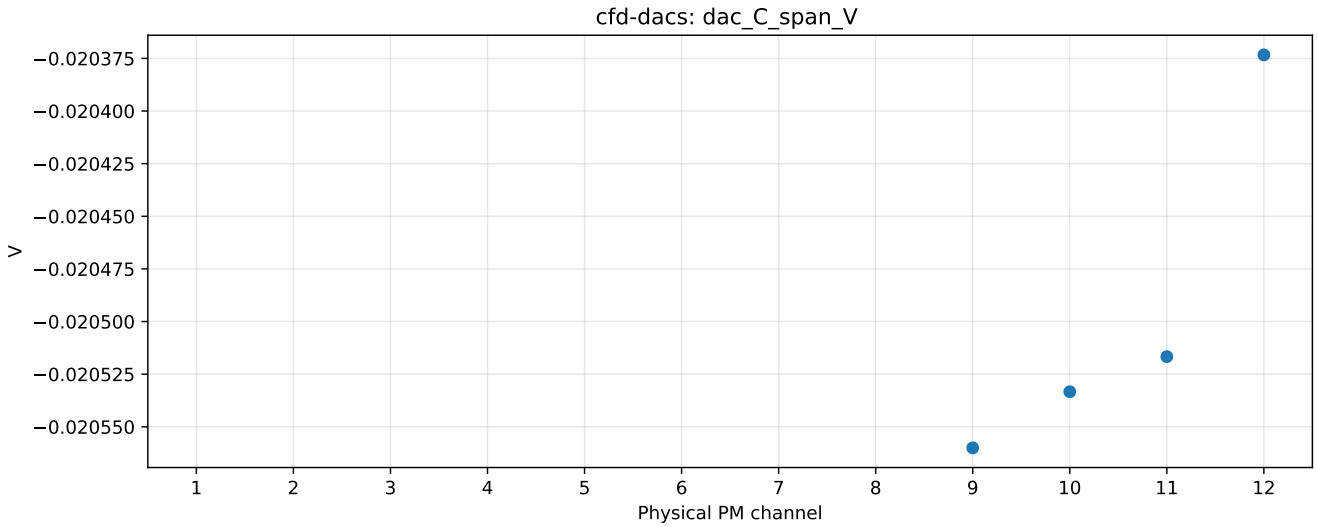

cfd-dacs: dac_B_maximum_reverse_step_V
0.04
0.02
0.00
−0.02
−0.04
V
1
2
3
4
5
6
7
8
9
10
11
12
Physical PM channel

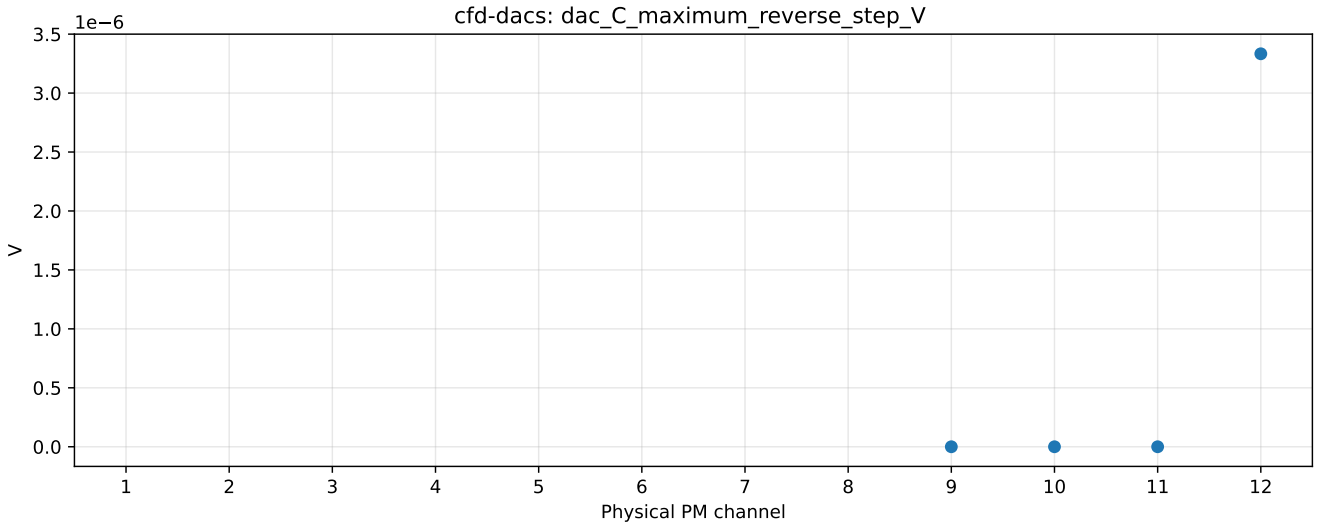

cfd-dacs: dac_B_maximum_repeat_std_V
1e−5
8.0
7.5
7.0
6.5
6.0
5.5
5.0
V
1
2
3
4
5
6
7
8
9
10
11
12
Physical PM channel

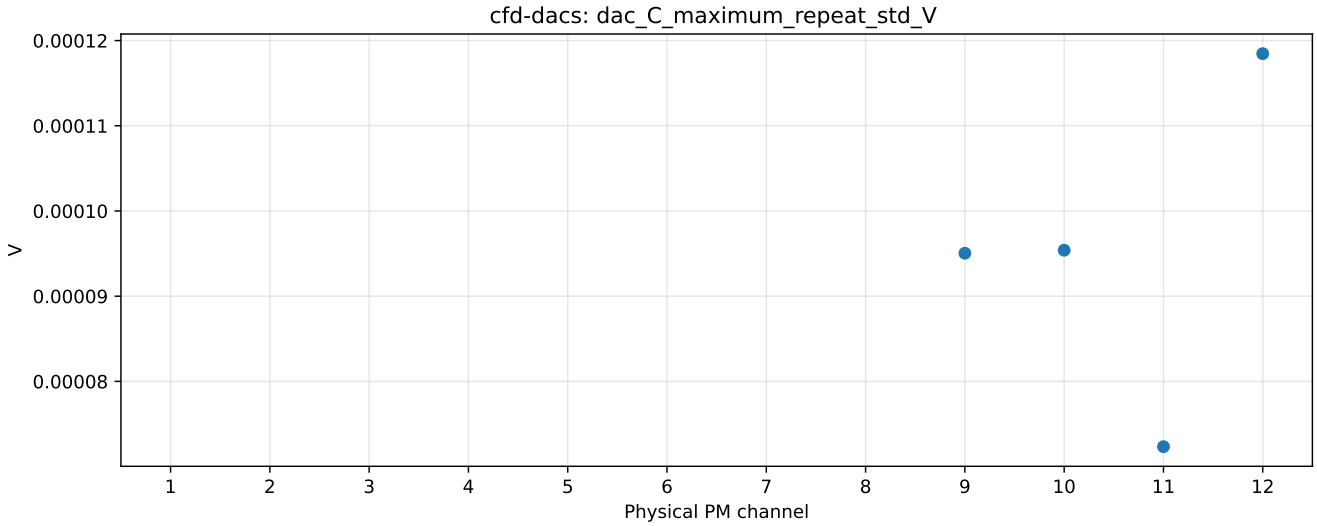

cfd-dacs: dac_B_interior_gain_error_percent
0.05
0.00
−0.05
−0.10
−0.15
%
1
2
3
4
5
6
7
8
9
10
11
12
Physical PM channel

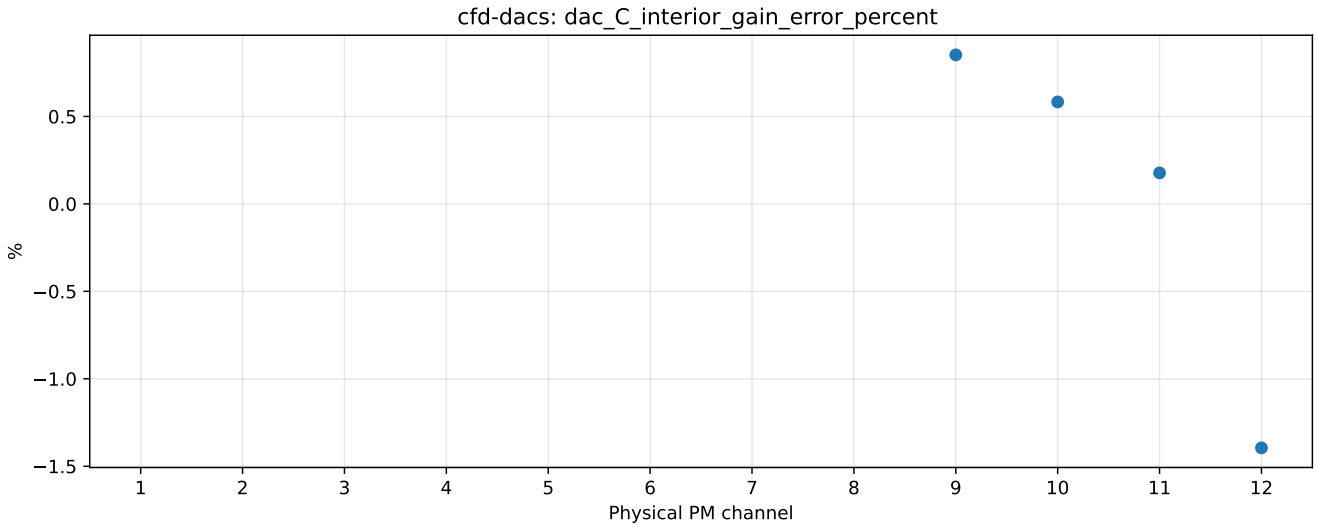

cfd-dacs: dac_C_minimum_code_V
+1.034
0.00095
0.00090
0.00085
0.00080
0.00075
0.00070
V
1
2
3
4
5
6
7
8
9
10
11
12
Physical PM channel

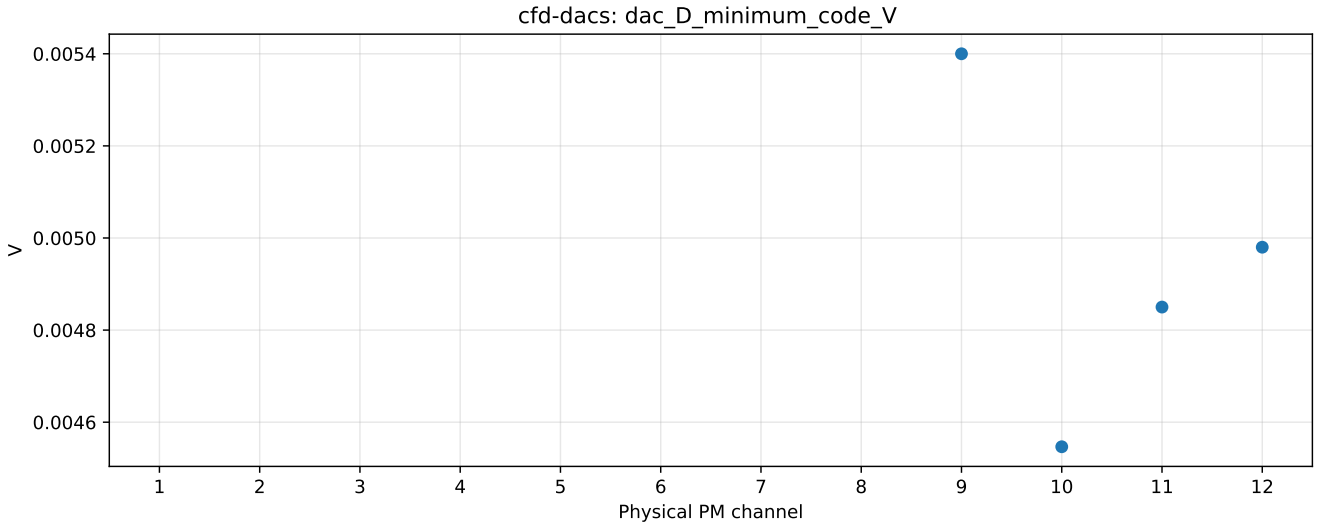

cfd-dacs: dac_C_maximum_code_V
+1.014
0.00040
0.00035
0.00030
0.00025
0.00020
0.00015
V
1
2
3
4
5
6
7
8
9
10
11
12
Physical PM channel

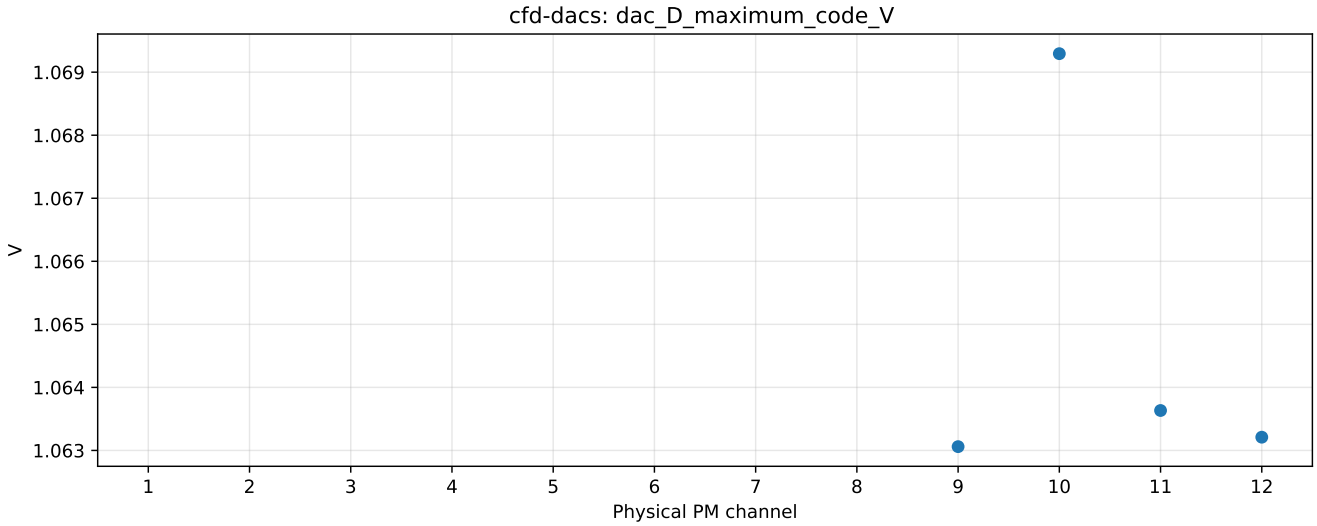

cfd-dacs: dac_C_span_V
−0.020375
−0.020400
−0.020425
−0.020450
−0.020475
−0.020500
−0.020525
−0.020550
V
1 2 3 4 5 6 7 8 9 10 11 12
Physical PM channel

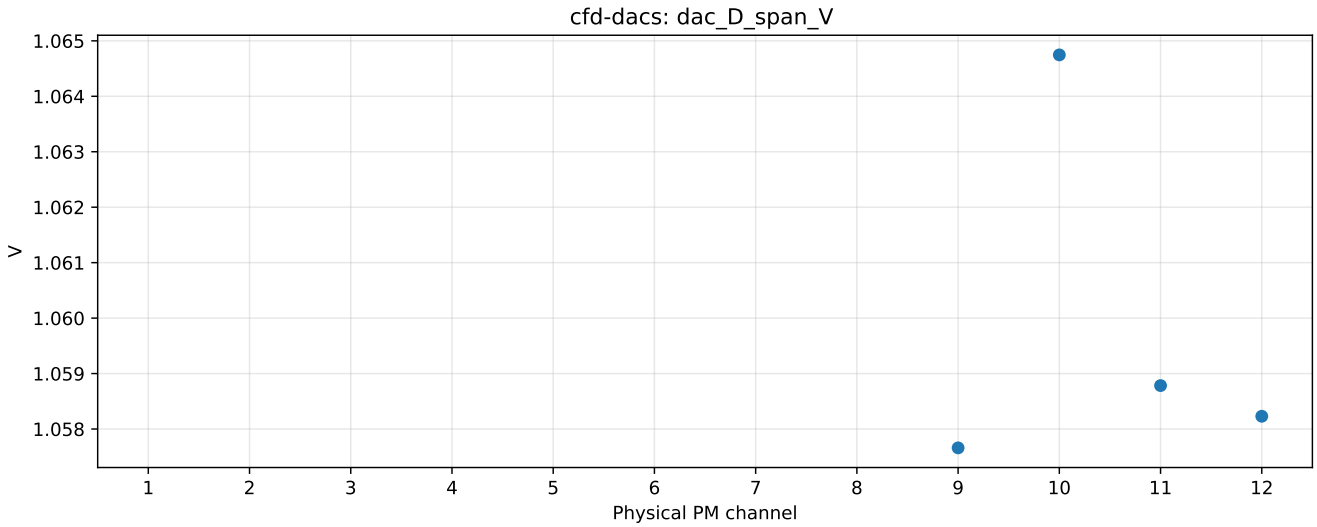

cfd-dacs: dac_C_maximum_reverse_step_V
1e−6
3.5
3.0
2.5
2.0
1.5
1.0
0.5
0.0
V
1 2 3 4 5 6 7 8 9 10 11 12
Physical PM channel

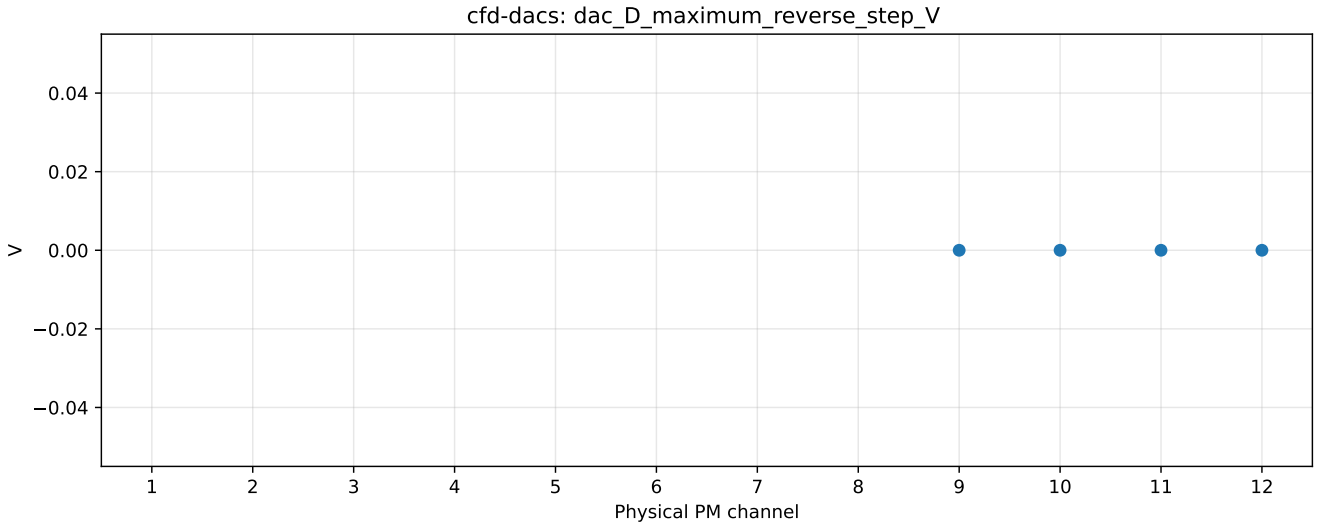

cfd-dacs: dac_C_maximum_repeat_std_V
0.00012
0.00011
0.00010
0.00009
0.00008
V
1 2 3 4 5 6 7 8 9 10 11 12
Physical PM channel

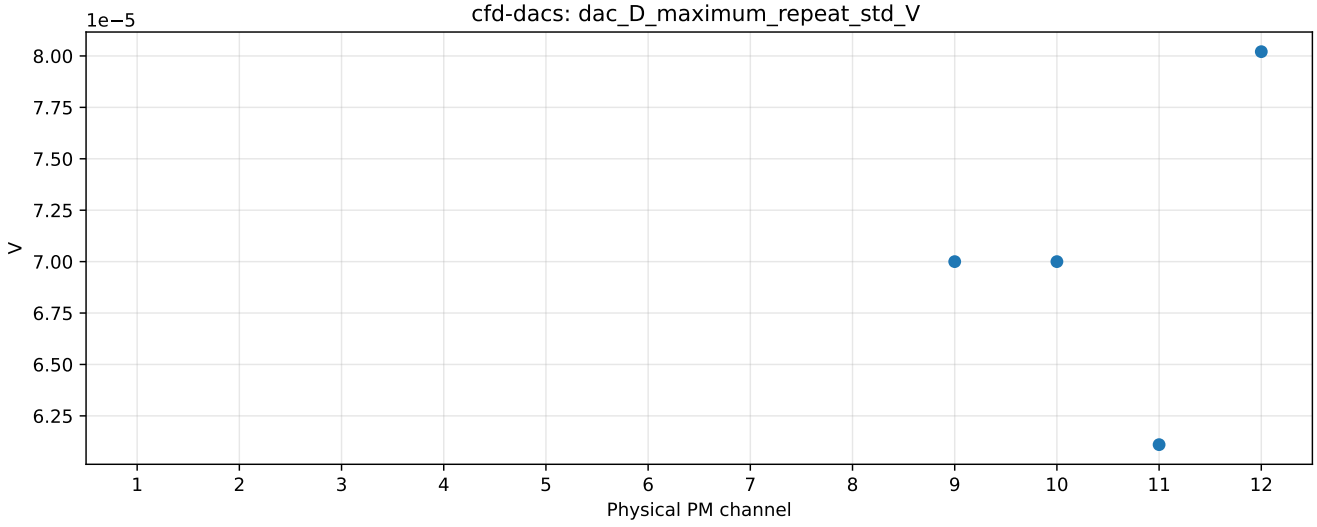

cfd-dacs: dac_C_interior_gain_error_percent
%
0.5
0.0
−0.5
−1.0
−1.5
1
2
3
4
5
6
7
8
9
10
11
12
Physical PM channel

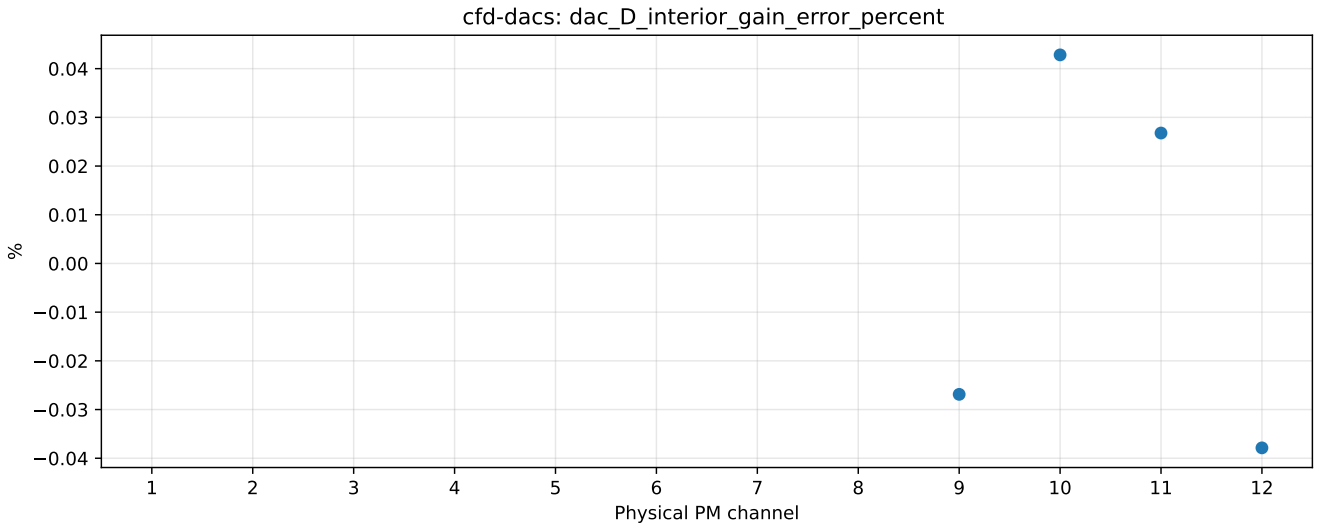

cfd-dacs: dac_D_minimum_code_V
V
0.0054
0.0052
0.0050
0.0048
0.0046
1
2
3
4
5
6
7
8
9
10
11
12
Physical PM channel

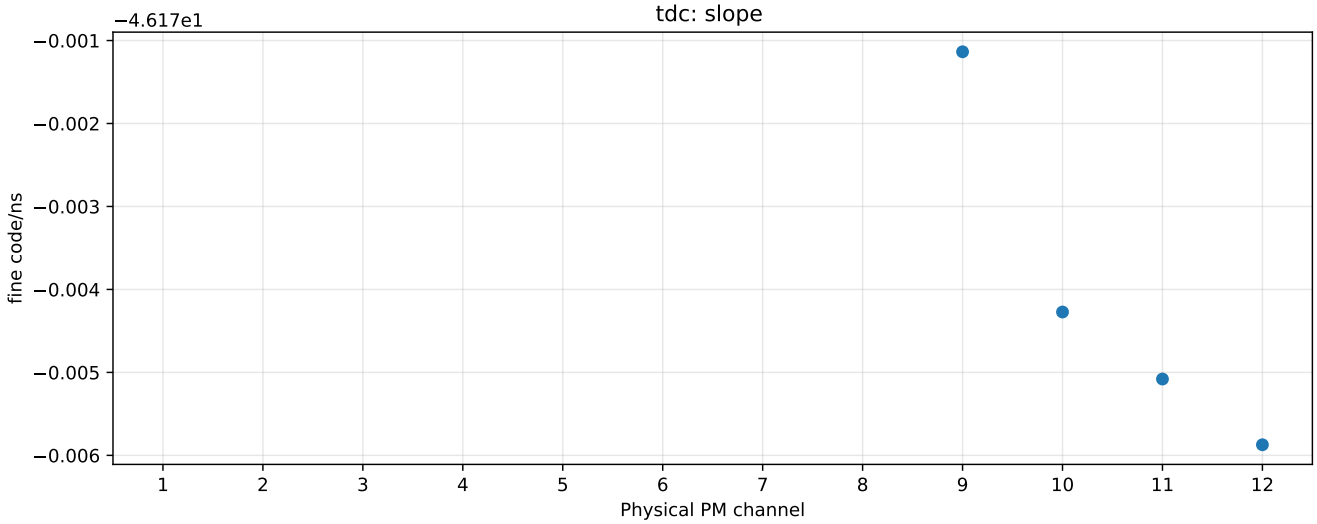

cfd-dacs: dac_D_maximum_code_V
V
1.069
1.068
1.067
1.066
1.065
1.064
1.063
1
2
3
4
5
6
7
8
9
10
11
12
Physical PM channel

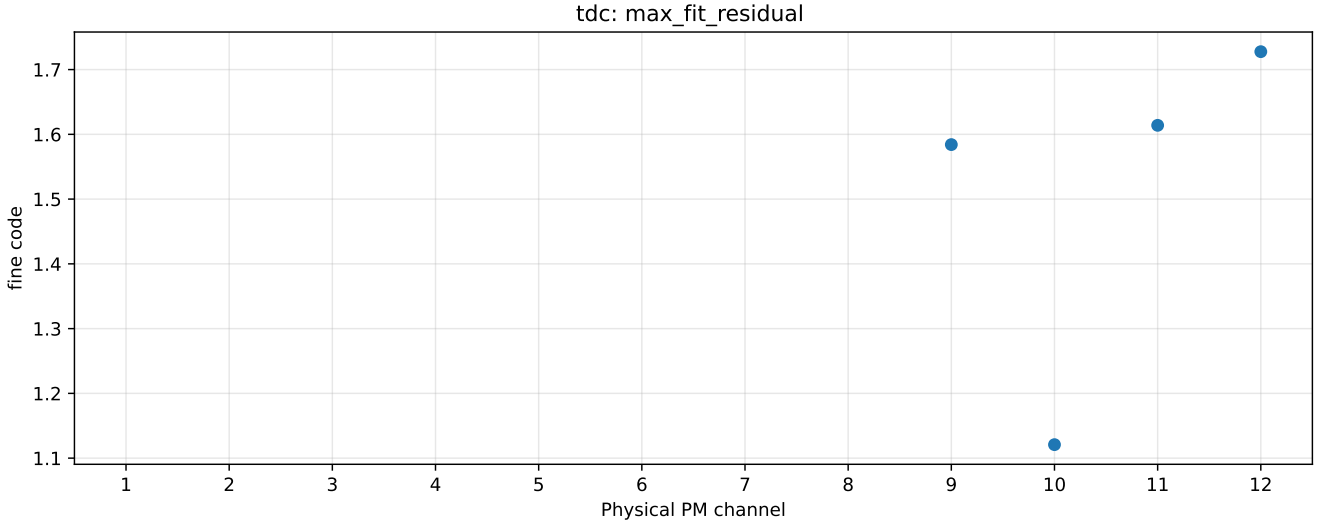

cfd-dacs: dac_D_span_V
1.065
1.064
1.063
1.062
1.061
1.060
1.059
1.058
V
1
2
3
4
5
6
7
8
9
10
11
12
Physical PM channel

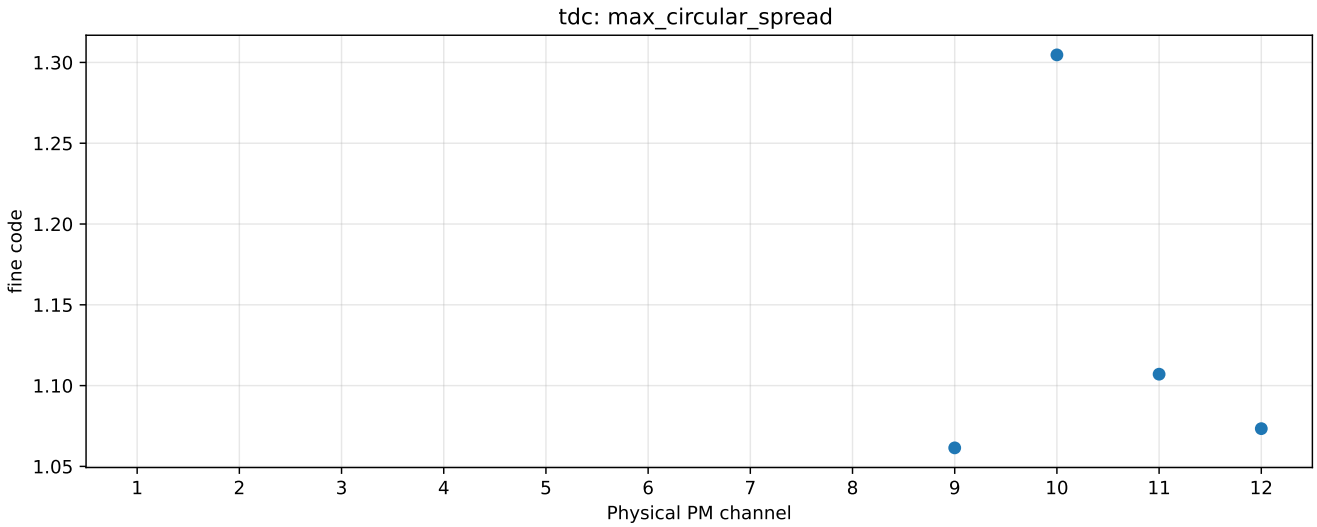

cfd-dacs: dac_D_maximum_reverse_step_V
0.04
0.02
0.00
−0.02
−0.04
V
1
2
3
4
5
6
7
8
9
10
11
12
Physical PM channel

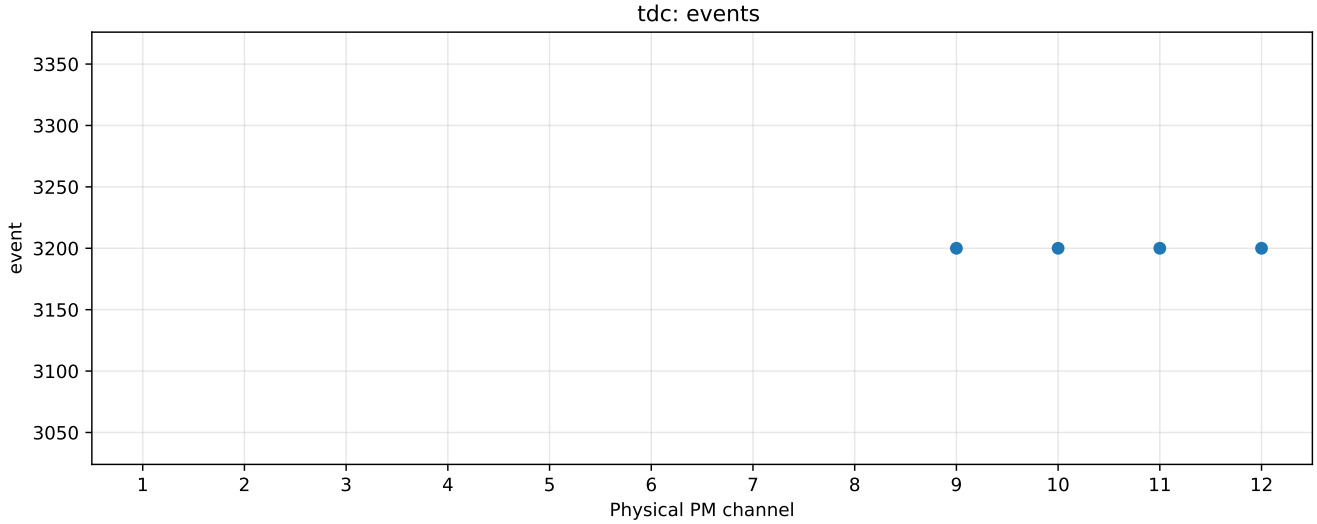

cfd-dacs: dac_D_maximum_repeat_std_V
1e−5
8.00
7.75
7.50
7.25
7.00
6.75
6.50
6.25
V
1
2
3
4
5
6
7
8
9
10
11
12
Physical PM channel

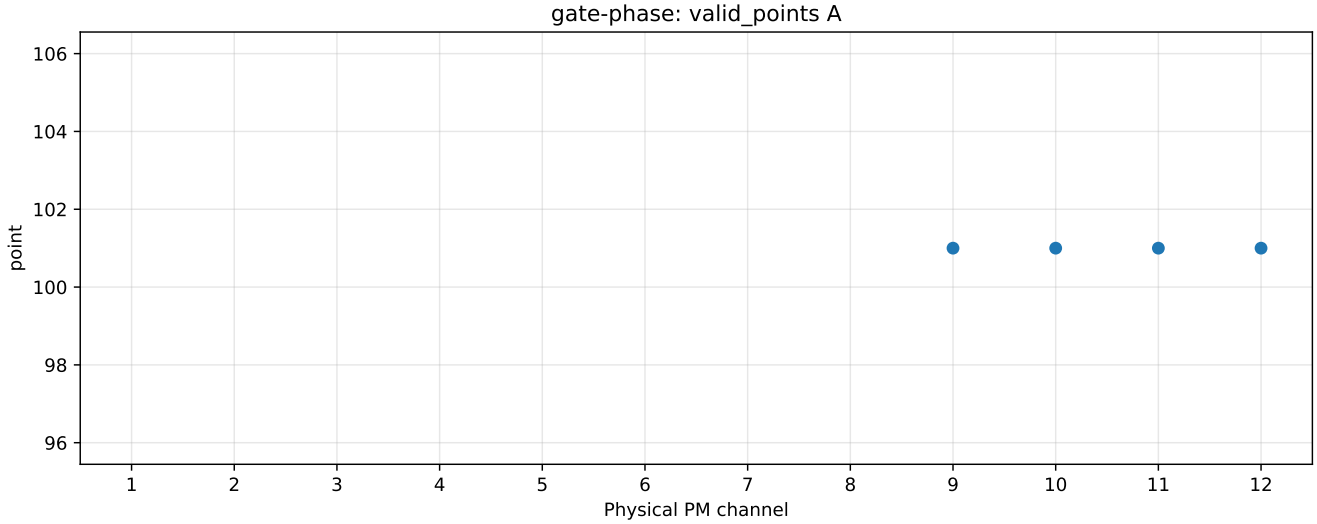

cfd-dacs: dac_D_interior_gain_error_percent
%
0.04
0.03
0.02
0.01
0.00
−0.01
−0.02
−0.03
−0.04
1
2
3
4
5
6
7
8
9
10
11
12
Physical PM channel

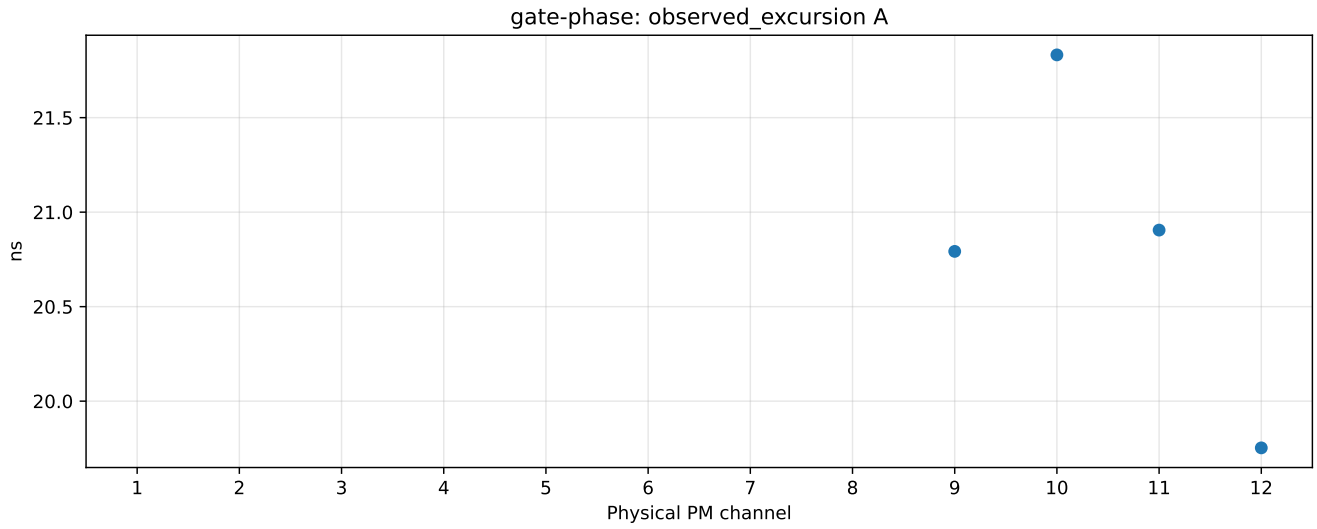

tdc: slope
−4.617e1
fine code/ns
−0.001
−0.002
−0.003
−0.004
−0.005
−0.006
1
2
3
4
5
6
7
8
9
10
11
12
Physical PM channel

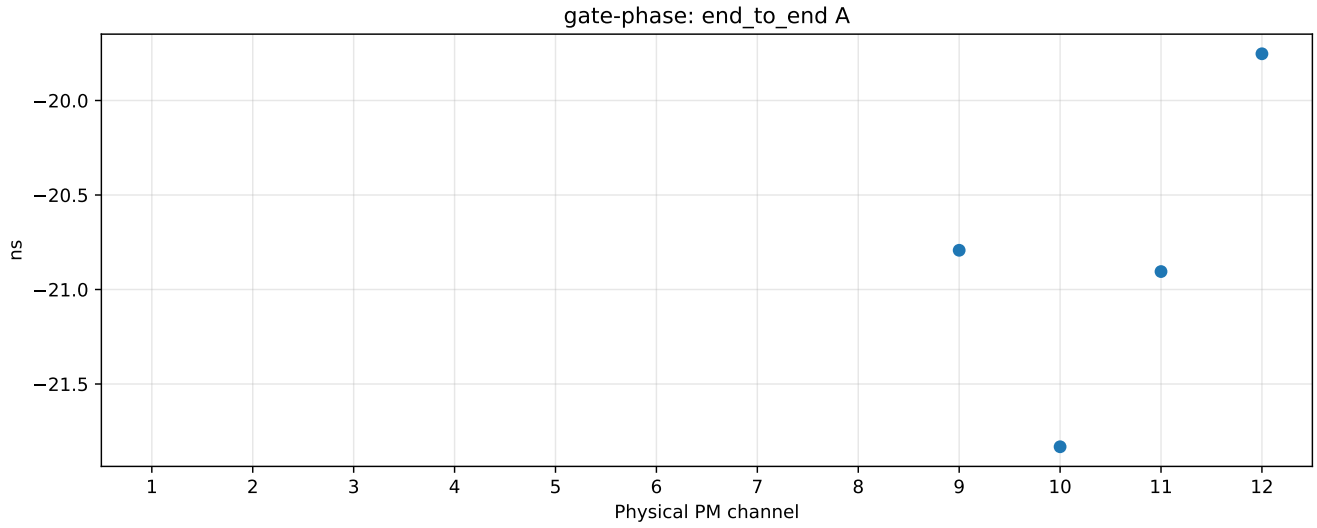

tdc: max_fit_residual
fine code
1.7
1.6
1.5
1.4
1.3
1.2
1.1
1
2
3
4
5
6
7
8
9
10
11
12
Physical PM channel

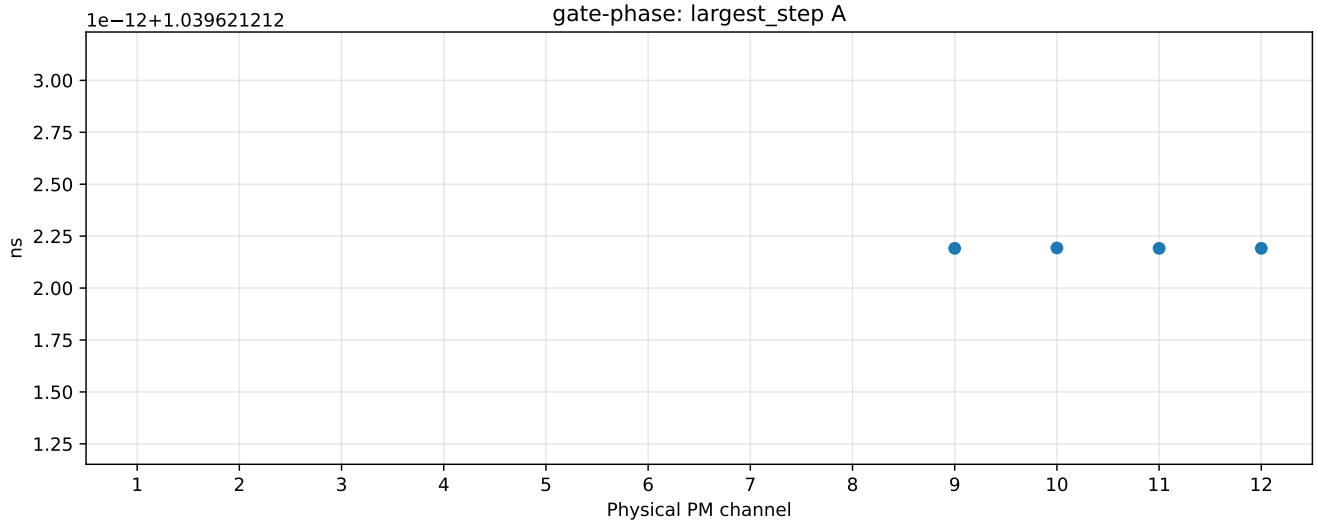

tdc: max_circular_spread
fine code
1.30
1.25
1.20
1.15
1.10
1.05
1 2 3 4 5 6 7 8 9 10 11 12
Physical PM channel

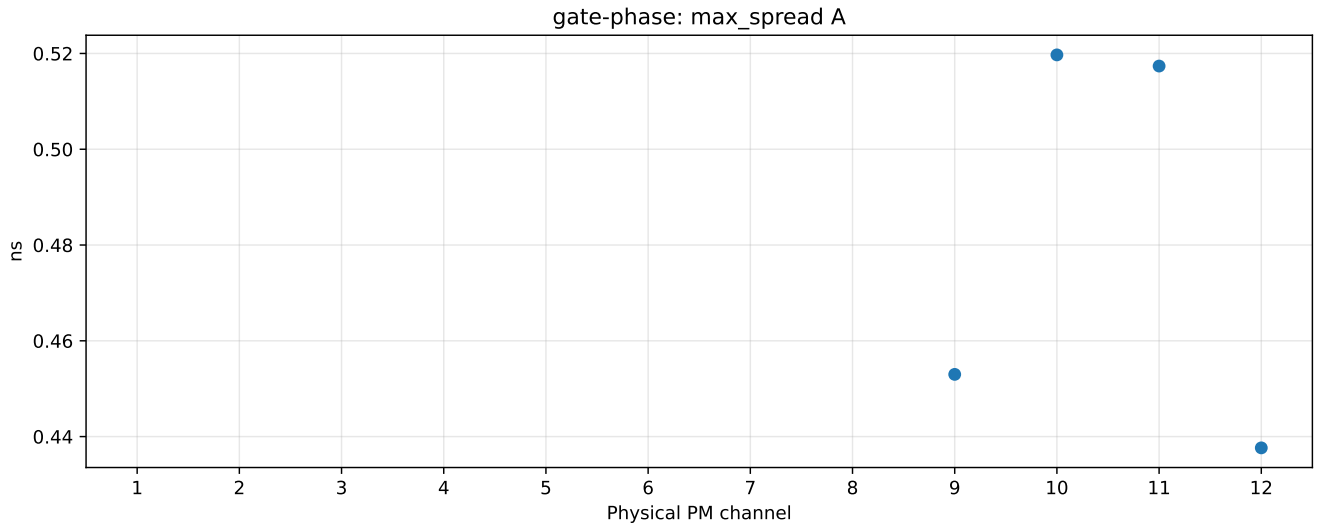

tdc: events
event
3350
3300
3250
3200
3150
3100
3050
1 2 3 4 5 6 7 8 9 10 11 12
Physical PM channel

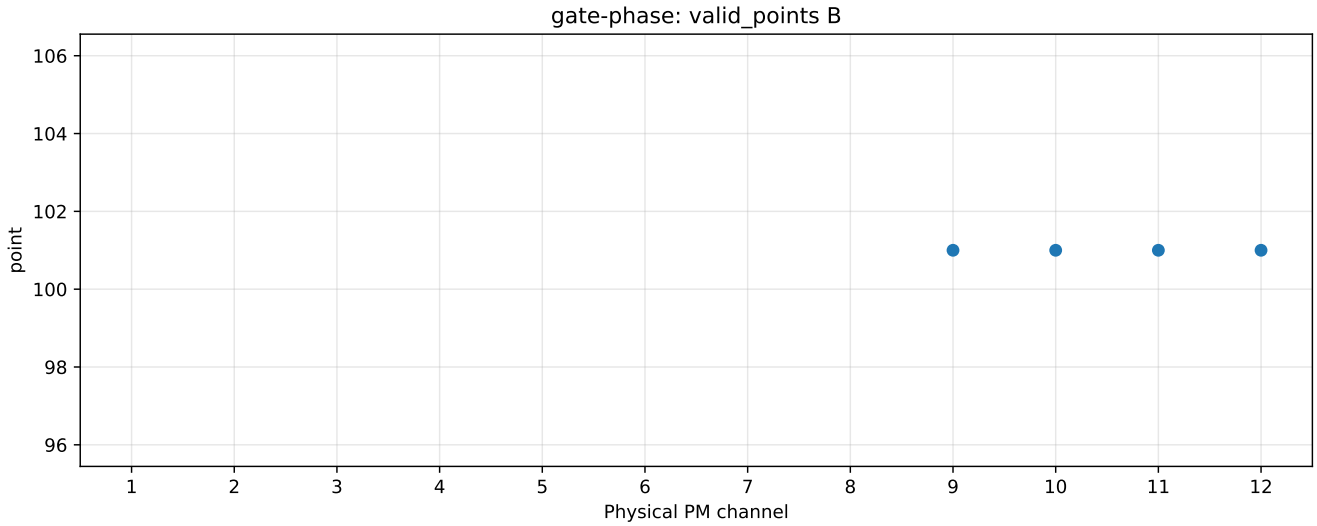

gate-phase: valid_points A
point
106
104
102
100
98
96
1 2 3 4 5 6 7 8 9 10 11 12
Physical PM channel

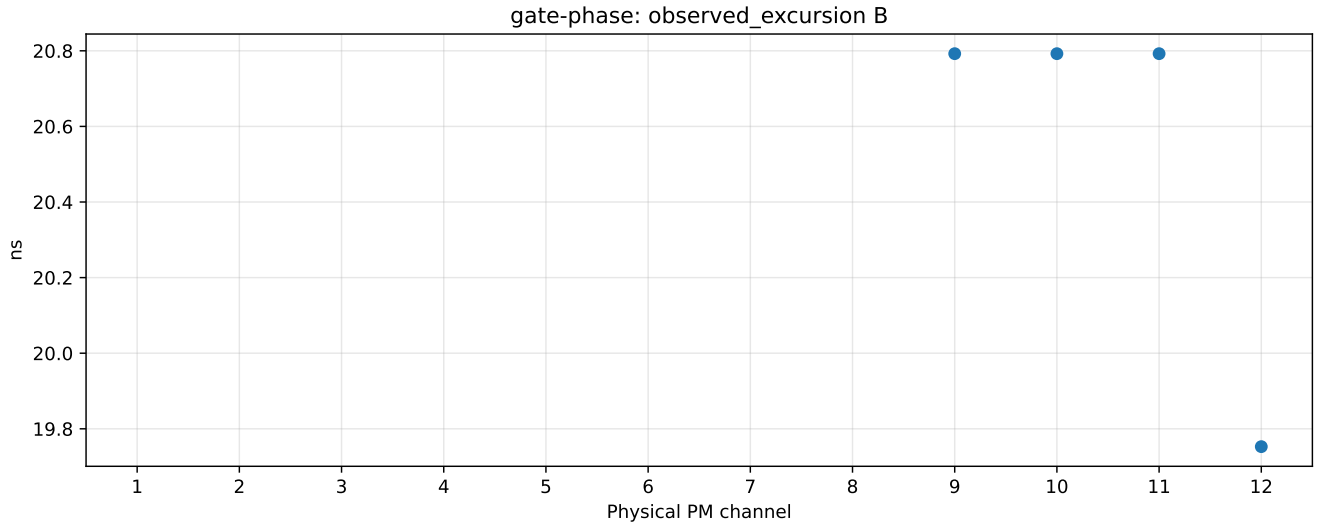

gate-phase: observed_excursion A
ns
21.5
21.0
20.5
20.0
1
2
3
4
5
6
7
8
9
10
11
12
Physical PM channel

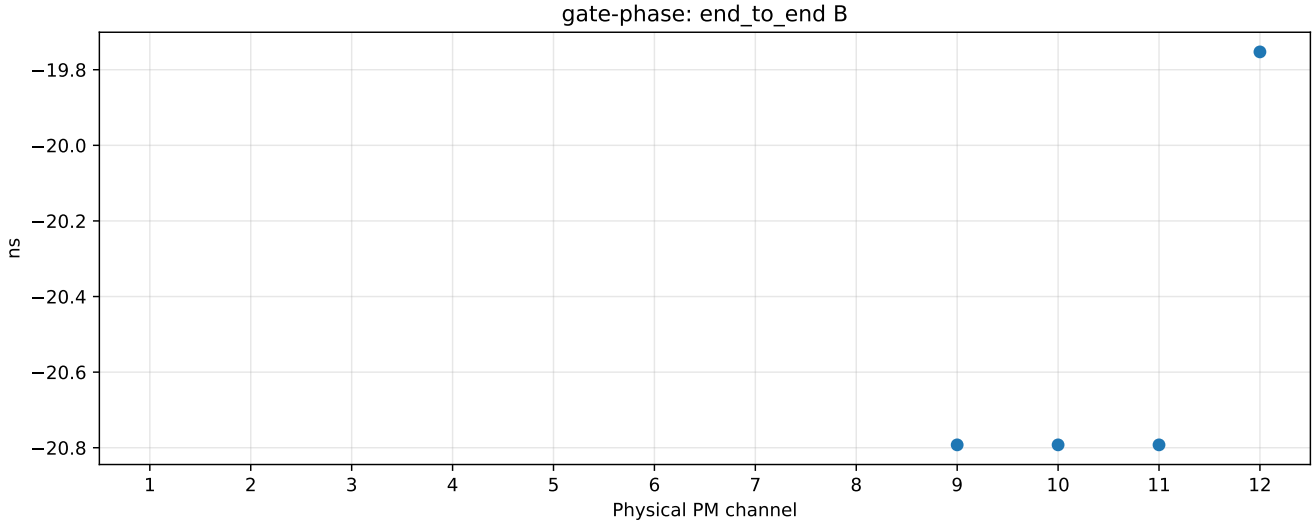

gate-phase: end_to_end A
ns
−20.0
−20.5
−21.0
−21.5
1
2
3
4
5
6
7
8
9
10
11
12
Physical PM channel

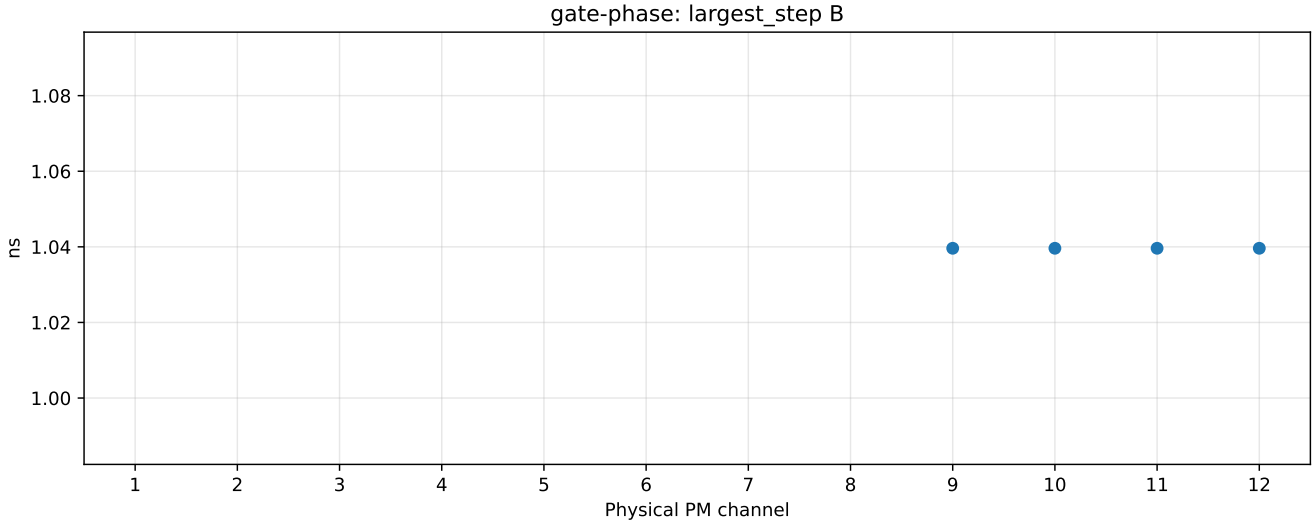

gate-phase: largest_step A
1e−12+1.039621212
ns
3.00
2.75
2.50
2.25
2.00
1.75
1.50
1.25
1
2
3
4
5
6
7
8
9
10
11
12
Physical PM channel

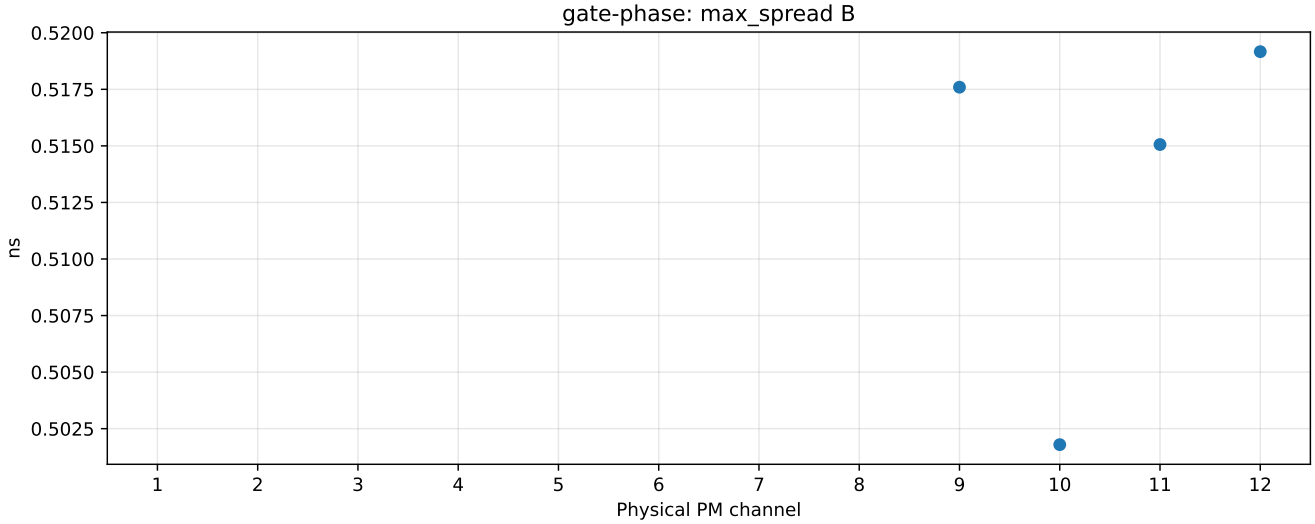

gate-phase: max_spread A
0.52
0.50
0.48
0.46
0.44
ns
1
2
3
4
5
6
7
8
9
10
11
12
Physical PM channel


gate-phase: valid_points B
106
104
102
100
98
96
point
1
2
3
4
5
6
7
8
9
10
11
12
Physical PM channel


gate-phase: observed_excursion B
20.8
20.6
20.4
20.2
20.0
19.8
ns
1
2
3
4
5
6
7
8
9
10
11
12
Physical PM channel


gate-phase: end_to_end B
ns
−19.8
−20.0
−20.2
−20.4
−20.6
−20.8
1
2
3
4
5
6
7
8
9
10
11
12
Physical PM channel


gate-phase: largest_step B
ns
1.08
1.06
1.04
1.02
1.00
1
2
3
4
5
6
7
8
9
10
11
12
Physical PM channel


gate-phase: max_spread B
ns
0.5200
0.5175
0.5150
0.5125
0.5100
0.5075
0.5050
0.5025
1
2
3
4
5
6
7
8
9
10
11
12
Physical PM channel